

Long–Short Portfolio Return Book

Cumulative return of each factor's long–short portfolio, net of 20 bp transaction cost.
225 factors over 199 months, ordered by in-sample CAGR.

HOW TO READ

Each curve starts at 0 % in January 2010 and compounds monthly long–short returns. The line is coloured by the factor's style; the figure at the right edge is the period-end cumulative return.

FACTOR STYLES

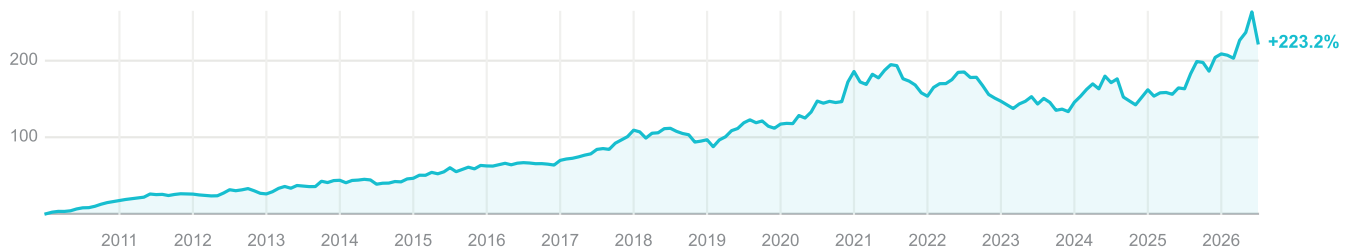
● Valuation	35	● Price Momentum	50
● Earnings Quality	46	● Size	4
● Analyst Expectations	18	● Historical Growth	38
● Capital Efficiency	27	● Volatility	7

001 In current model portfolio (28 factors)

001 ● 3M Revision in FY2 EPS Estimate

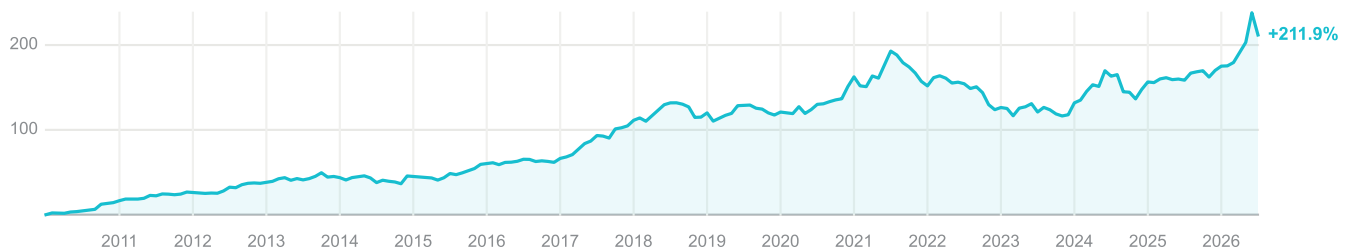
Analyst Expectations

CAGR +7.4%

**002 ● Street Revision Magnitude**

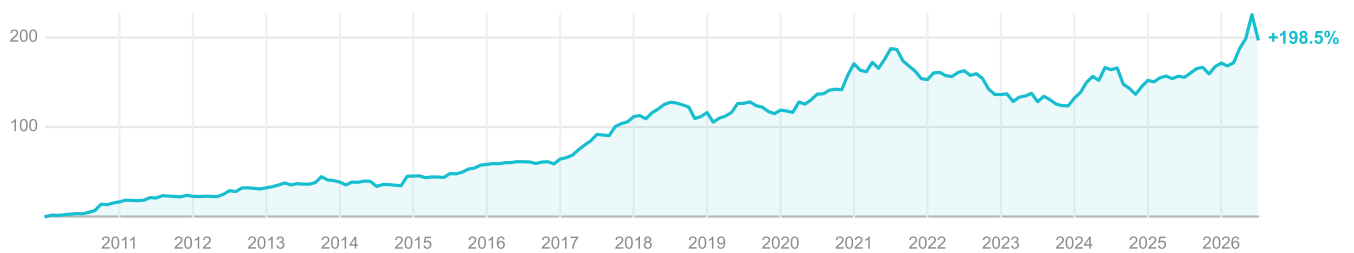
Analyst Expectations

CAGR +7.1%

**003 ● 3M Revision in FY1 EPS Estimate**

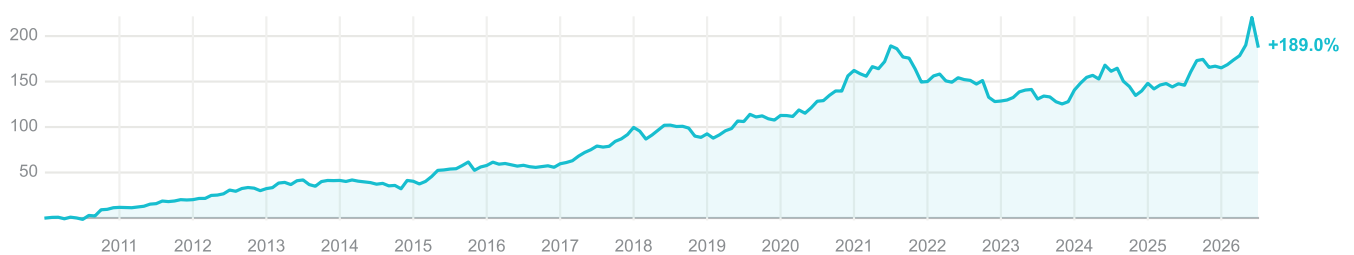
Analyst Expectations

CAGR +6.9%

**004 ● Number of EPS FY1 Revisions**

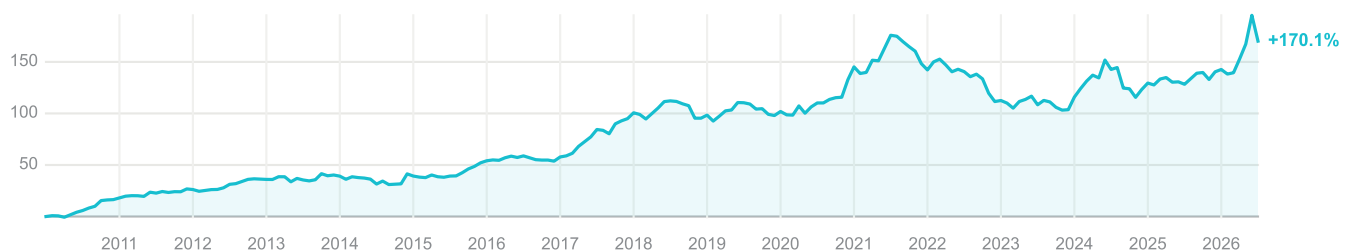
Analyst Expectations

CAGR +6.6%

**005 ● Adjusted Revision Magnitude**

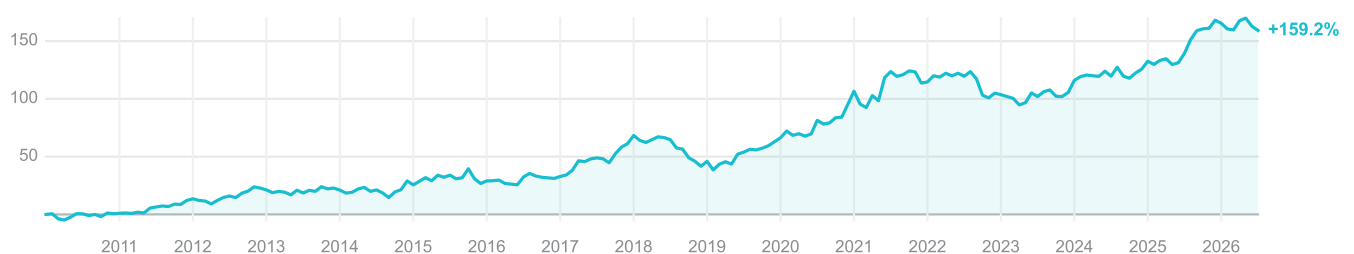
Analyst Expectations

CAGR +6.2%

**006 ● Standardized Unexpected Earnings**

Analyst Expectations

CAGR +5.9%



007 ● 6M Chg in Target Price

Analyst Expectations

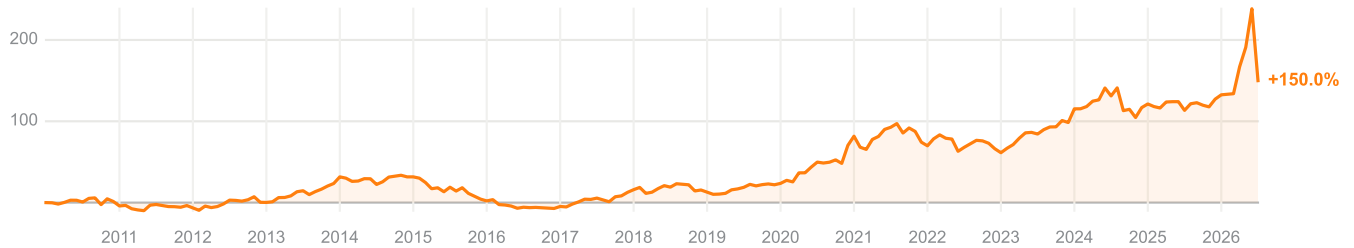
CAGR +5.8%



008 ● Slope of 52W Price Trend

Price Momentum

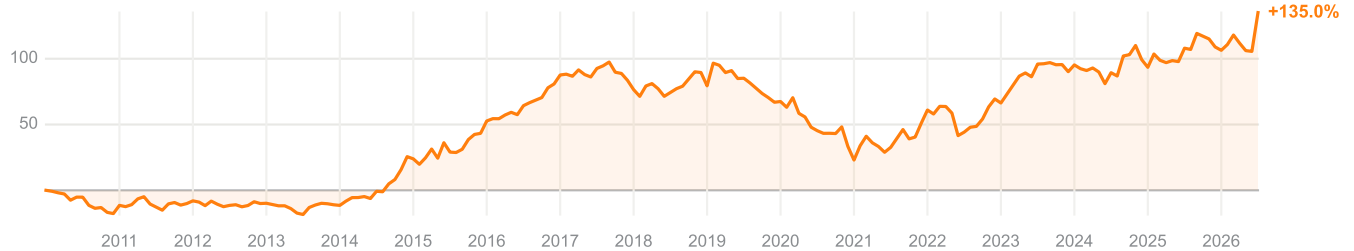
CAGR +5.7%



009 ● Sharpe Ratio

Price Momentum

CAGR +5.3%



010 ● Cash Flow Return on Invested Capital

Capital Efficiency

CAGR +5.2%



011 ● Operating Cash Flow to Enterprise Value

Valuation

CAGR +5.0%



012 ● Closing Price to 260 Day Low

Price Momentum

CAGR +5.0%



013 ● Dividend per Share Growth

Historical Growth

CAGR +5.0%

**014 ● 24M Residual Return Variance**

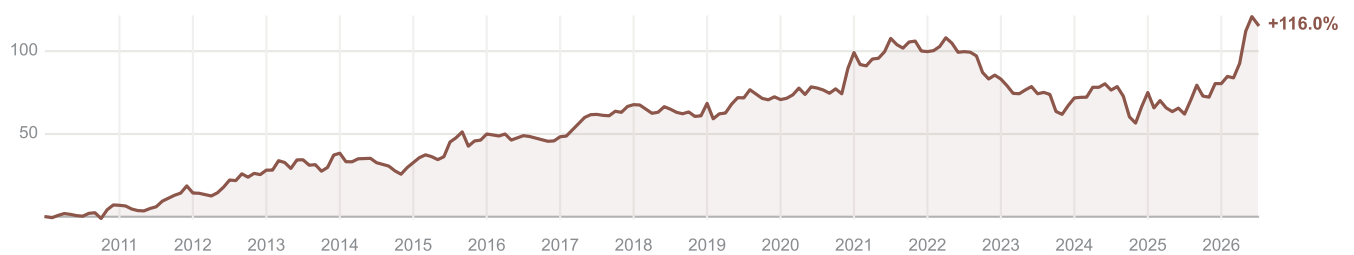
Price Momentum

CAGR +5.0%

**015 ● 3Y Chg in Earnings per Share**

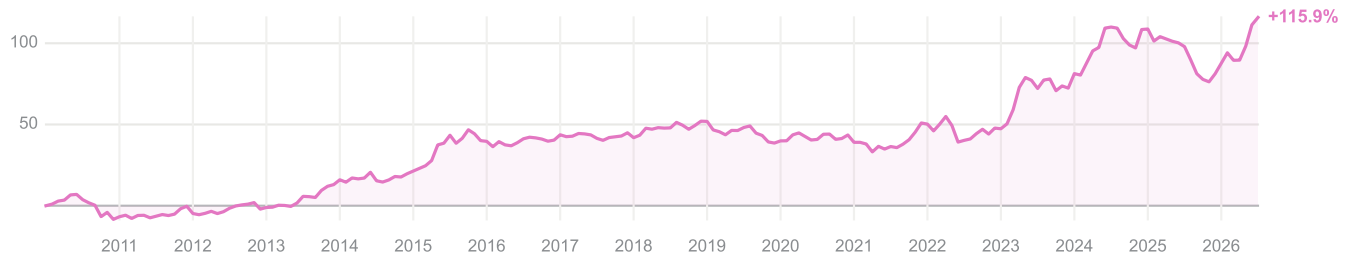
Historical Growth

CAGR +4.8%

**016 ● ROE 20 Qtr Standard Deviation**

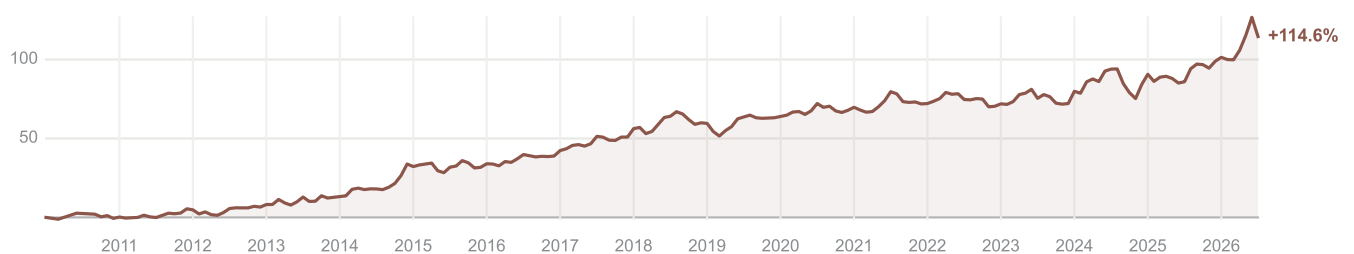
Earnings Quality

CAGR +4.8%

**017 ● 1Y Chg in Sales Turnover**

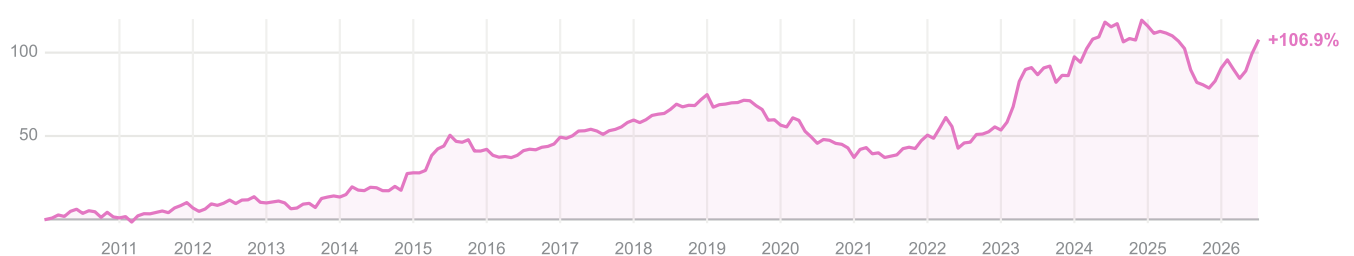
Historical Growth

CAGR +4.7%

**018 ● ROA 20 Qtr Standard Deviation**

Earnings Quality

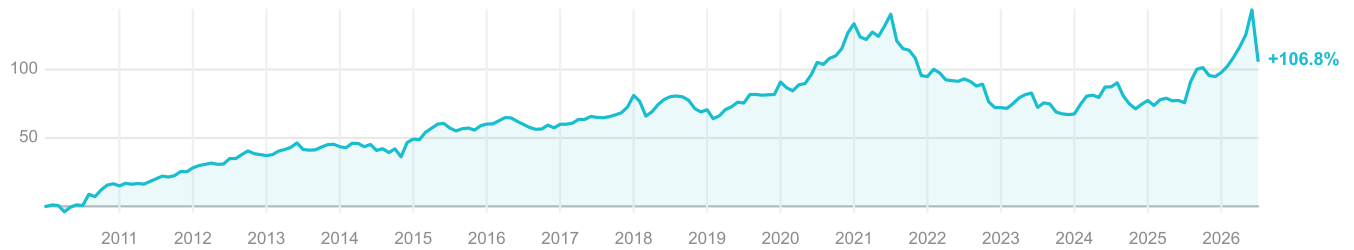
CAGR +4.5%



019 ● **Adjusted Number of EPS FY1 Revisions**

Analyst Expectations

CAGR +4.5%

020 ● **Operating Cash Flow to Equity**

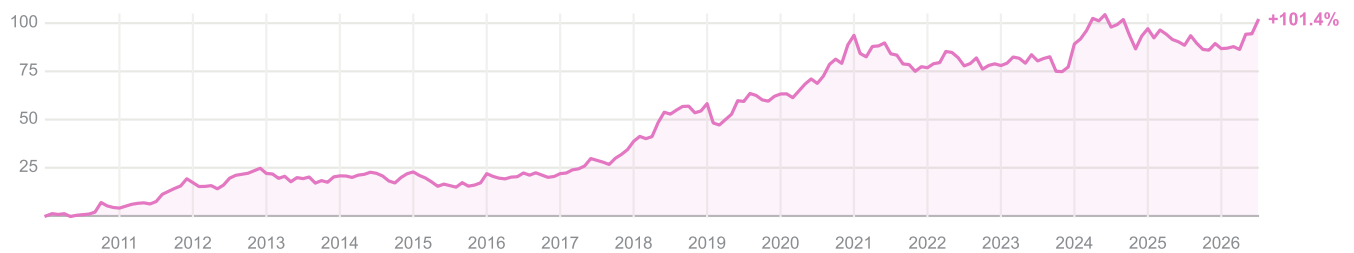
Capital Efficiency

CAGR +4.4%

021 ● **Free Cash Flow to Sales**

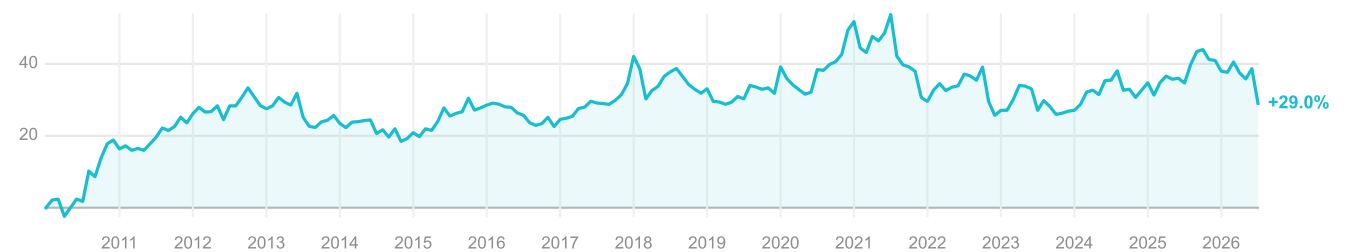
Earnings Quality

CAGR +4.3%

022 ● **Analyst Earnings Estimate Diffusion**

Analyst Expectations

CAGR +4.3%

023 ● **Operating Cash Flow Ratio**

Capital Efficiency

CAGR +4.3%

024 ● **Sales to Price Ratio**

Valuation

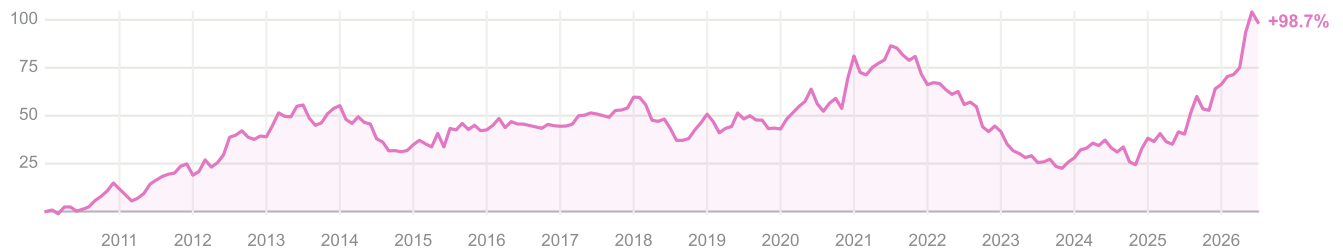
CAGR +4.3%



025 ● **Net Profit Margin**

Earnings Quality

CAGR +4.2%

026 ● **Long Term Debt to Equity Ratio**

Capital Efficiency

CAGR +4.2%

027 ● **1Y Chg in Operating Cash Flow per Share**

Historical Growth

CAGR +4.2%

028 ● **Free Cash Flow to Common Equity**

Valuation

CAGR +3.9%

029 ● **Tobin's Q Ratio**

Capital Efficiency

CAGR +3.9%

030 ● **Operating Cash Flow to Price**

Valuation

CAGR +3.8%



031 ● **Current Liability to Price**

Valuation

CAGR +3.8%

032 ● **Gross Profit to Assets**

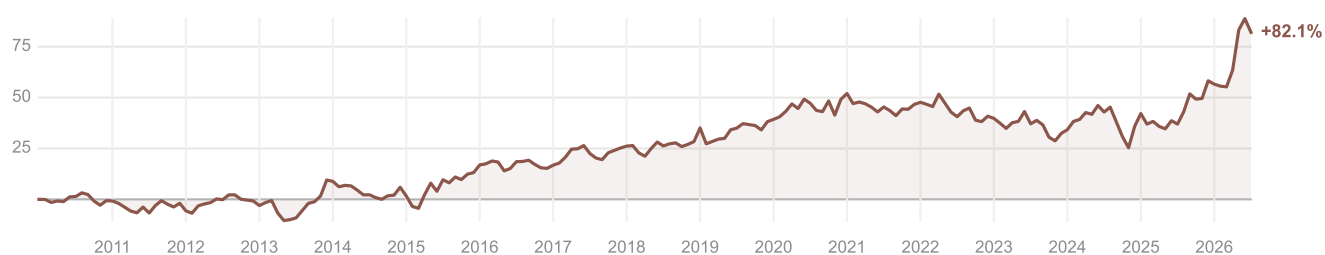
Capital Efficiency

CAGR +3.7%

033 ● **3Y Chg in Operating Margin**

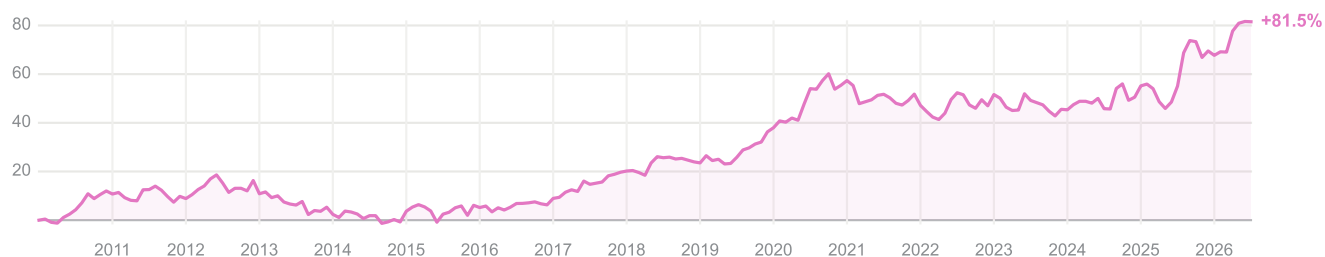
Historical Growth

CAGR +3.7%

034 ● **Asset Turnover**

Earnings Quality

CAGR +3.7%

035 ● **180D Price TStat**

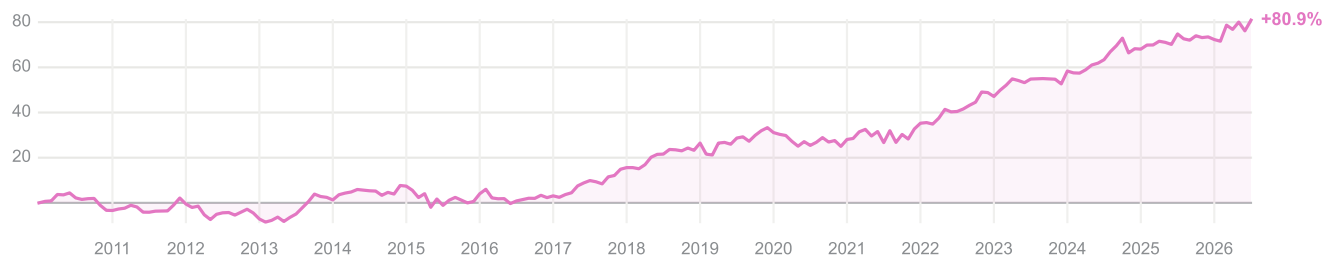
Price Momentum

CAGR +3.7%

036 ● **Adjusted Accruals**

Earnings Quality

CAGR +3.7%



037 ● 3Y Chg in Operating Cash Flow per Share

Historical Growth

CAGR +3.6%



038 ● Institution Ownership Concentration

Price Momentum

CAGR +3.6%



039 ● Target Price Gap to 6M EMA

Analyst Expectations

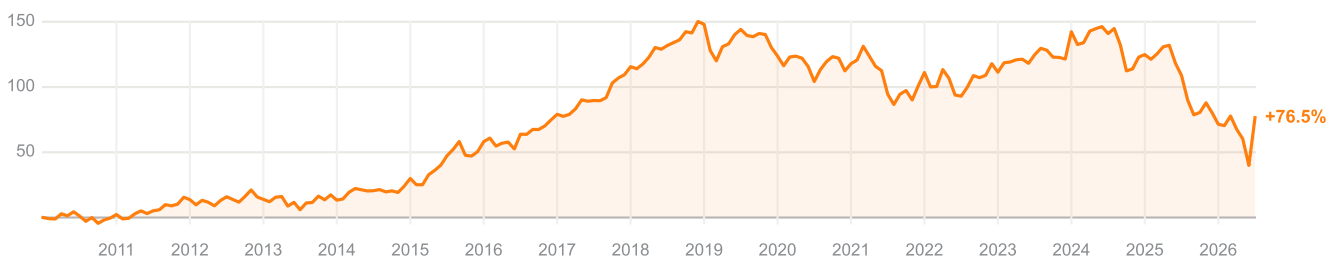
CAGR +3.5%



040 ● Share Turnover

Price Momentum

CAGR +3.5%



041 ● 1Y Chg in Sales to Price

Historical Growth

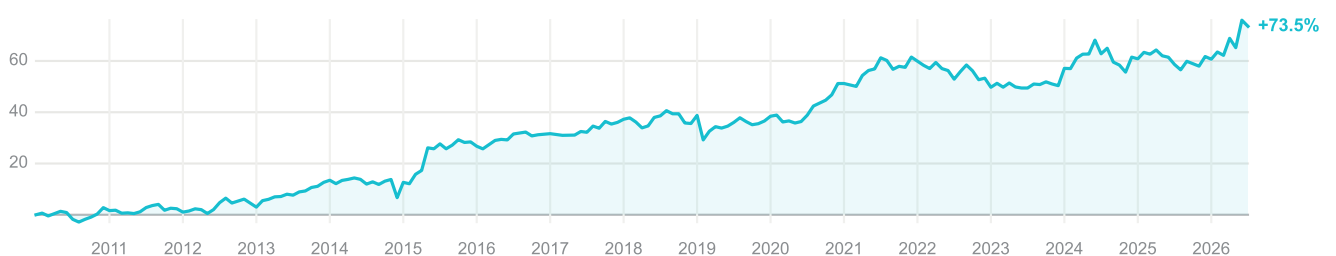
CAGR +3.5%



042 ● 6M Avg Chg 1M Recommendation

Analyst Expectations

CAGR +3.4%



043 ● 1Y Chg in Asset Adjusted EPS

Historical Growth

CAGR +3.3%



044 ● 3Y Chg in Free Cash Flow per Share

Historical Growth

CAGR +3.3%



045 ● Capital Acquisition Ratio

Capital Efficiency

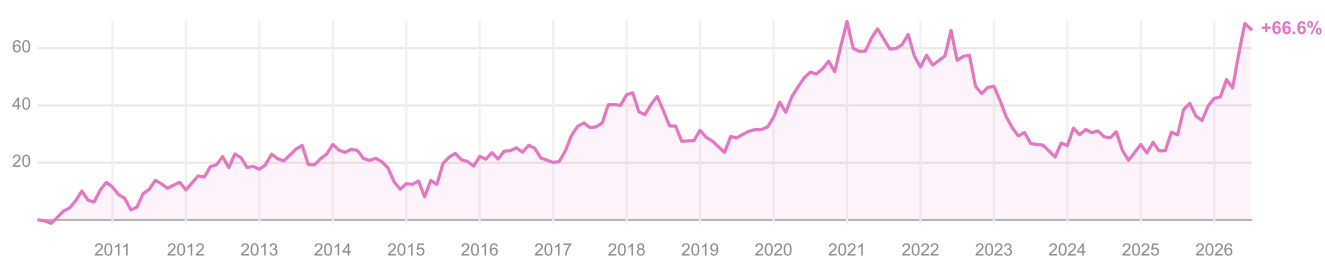
CAGR +3.2%



046 ● EBIT Margin

Earnings Quality

CAGR +3.1%



047 ● 1Y Chg in Asset Adjusted Free Cash Flow

Historical Growth

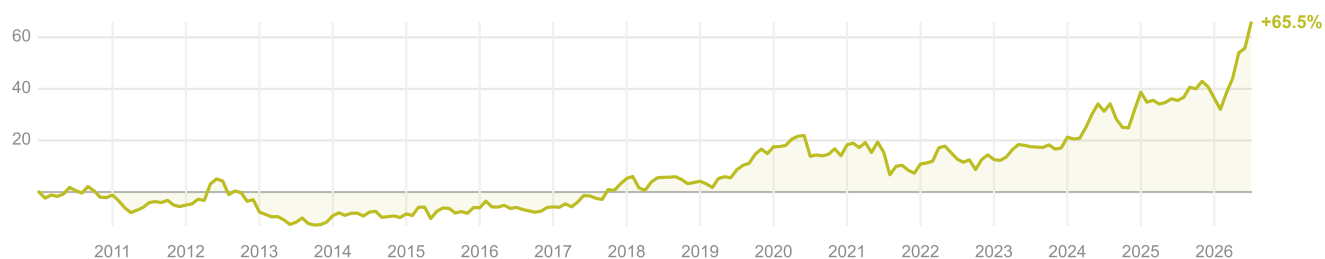
CAGR +3.1%



048 ● Receivables Turnover Ratio

Capital Efficiency

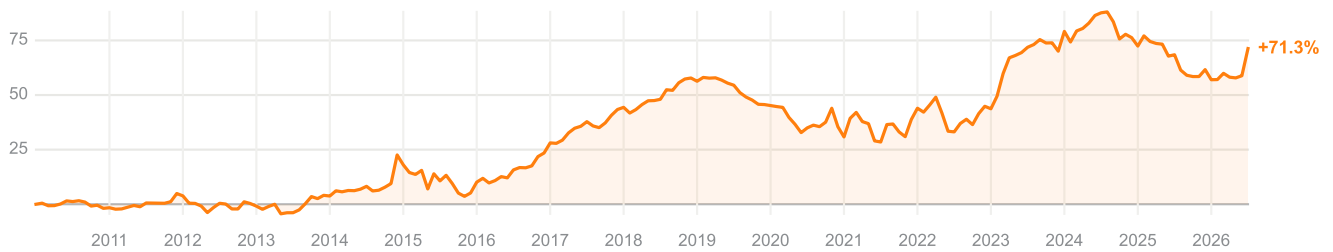
CAGR +3.1%



049 ● **Ownership Breadth Stability of Institution Ownership by Holders**

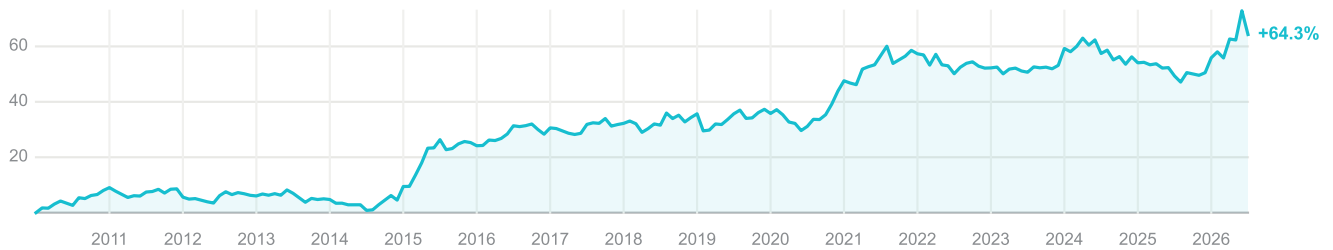
Price Momentum

CAGR +3.1%

050 ● **Buy-to-Sell Recommendation Ratio Less 3M EMA**

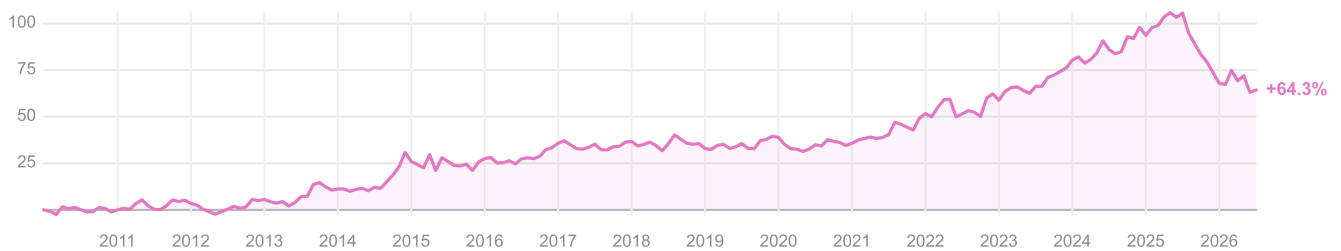
Analyst Expectations

CAGR +3.1%

051 ● **Working Capital Accruals**

Earnings Quality

CAGR +3.1%

052 ● **Assets to Price Ratio**

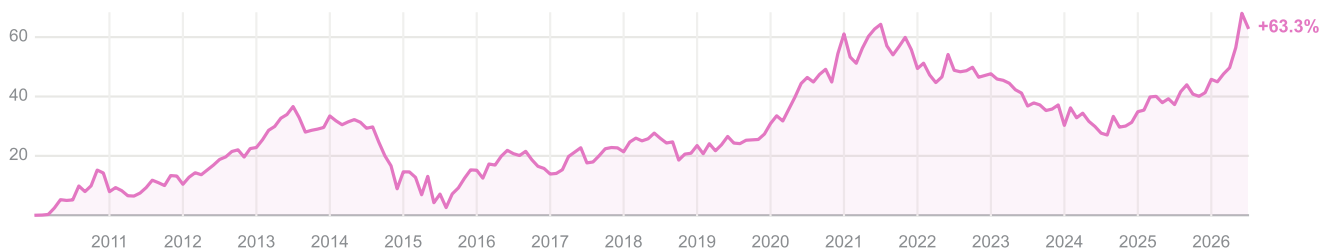
Valuation

CAGR +3.0%

053 ● **Gross Profit Margin**

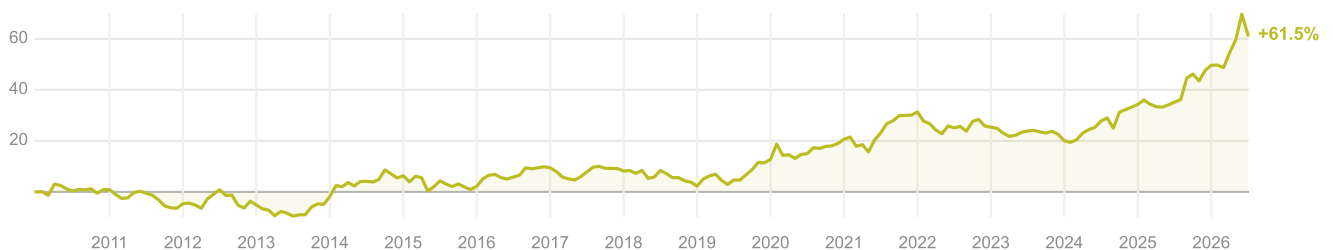
Earnings Quality

CAGR +3.0%

054 ● **Adjusted Interest Coverage Ratio**

Capital Efficiency

CAGR +2.9%



055 ● 1Y Chg in Cash Flow per Share

Historical Growth

CAGR +2.9%



056 ● 1Y Chg in Free Cash Flow per Share

Historical Growth

CAGR +2.9%



057 ● 1Y Chg in Cash Flow to Price

Historical Growth

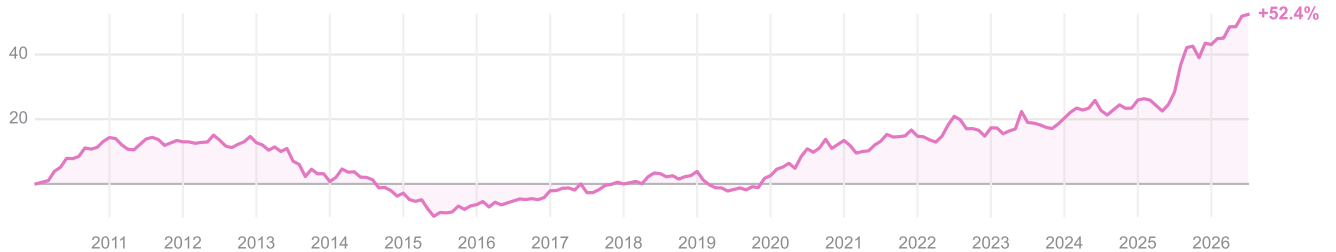
CAGR +2.9%



058 ● Sales to Invested Capital

Earnings Quality

CAGR +2.9%



059 ● 1Y Chg in Earnings to Price

Historical Growth

CAGR +2.9%



060 ● 6M Chg in Target Price Gap

Analyst Expectations

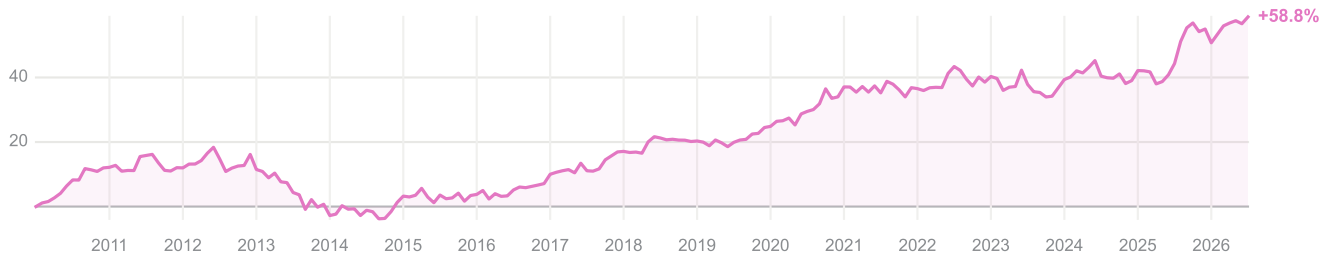
CAGR +2.9%



061 Equity Turnover

Earnings Quality

CAGR +2.8%

**062 Net Cash Flow to Price**

Valuation

CAGR +2.8%

**063 Cash to Price**

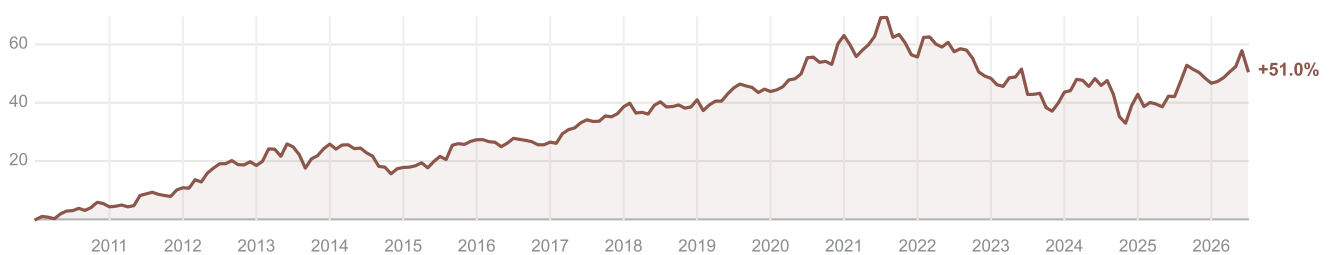
Valuation

CAGR +2.7%

**064 1Y Chg in Operating Margin**

Historical Growth

CAGR +2.7%

**065 Cash Flow to Equity**

Valuation

CAGR +2.7%

**066 Change EPS to Sales**

Earnings Quality

CAGR +2.7%



067 ● **Growth Flow to Price**

Valuation

CAGR +2.7%

068 ● **1Y Chg in Asset Adjusted Cash Flow**

Historical Growth

CAGR +2.7%

069 ● **1Y Chg in Earnings per Share**

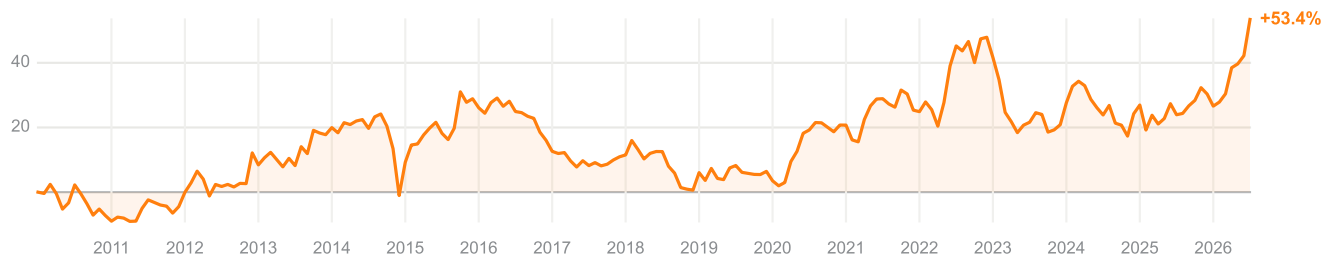
Historical Growth

CAGR +2.6%

070 ● **Risk Adjusted Relative Strength**

Price Momentum

CAGR +2.6%

071 ● **Free Cash Flow to Price**

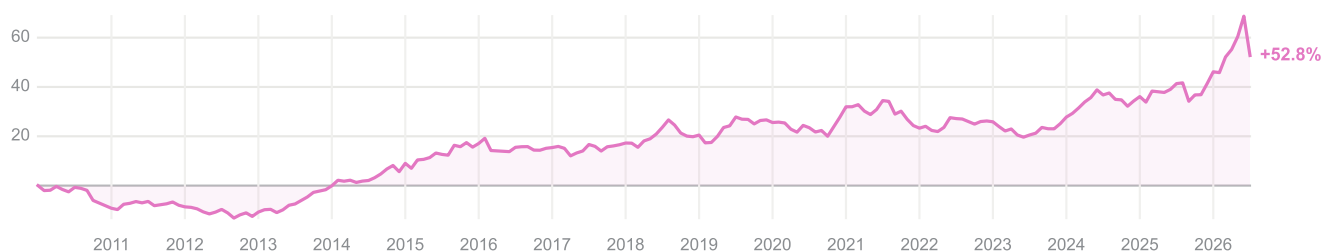
Valuation

CAGR +2.6%

072 ● **Unexpected Receivables Change**

Earnings Quality

CAGR +2.6%



073 ● 1Y Chg in Asset Adjusted Operating Cash Flow

Historical Growth

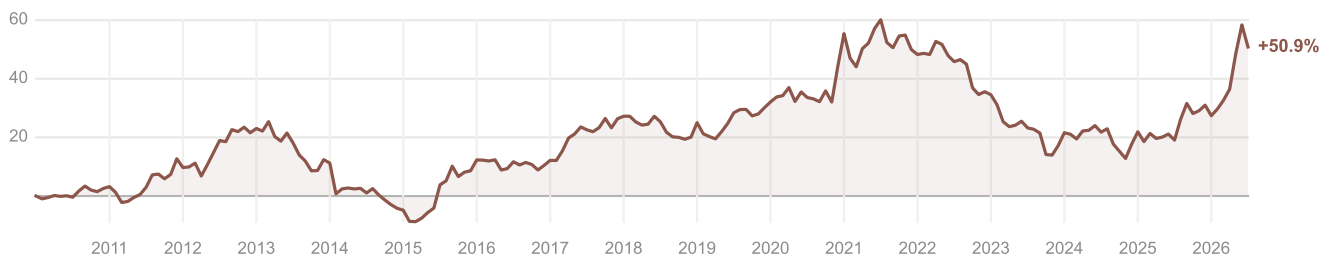
CAGR +2.6%



074 ● 3Y Chg in Asset Adjusted Cash Flow

Historical Growth

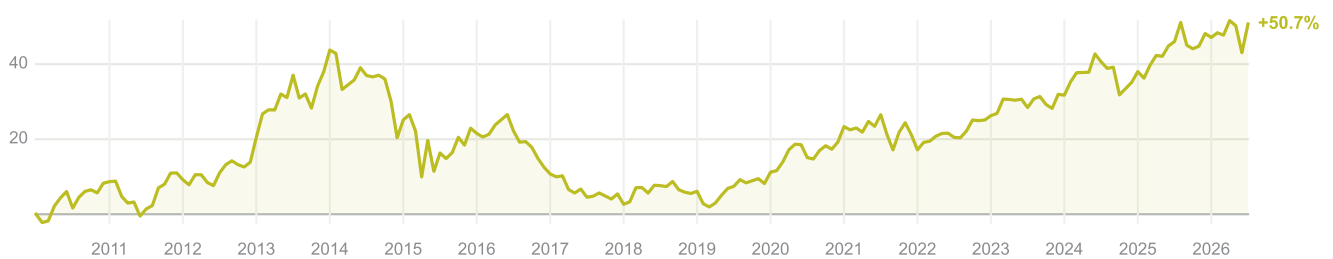
CAGR +2.5%



075 ● Long Term Debt to Assets Ratio

Capital Efficiency

CAGR +2.5%



076 ● 3Y Chg in Asset Adjusted Operating Cash Flow

Historical Growth

CAGR +2.5%



077 ● Adjusted EBIT to Price

Valuation

CAGR +2.4%



078 ● 3Y Average Annual Earnings Growth

Historical Growth

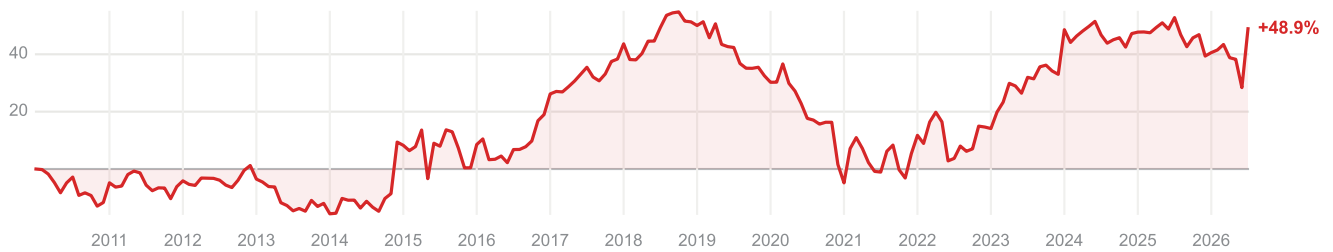
CAGR +2.4%



079 ● Sales to EV Ratio

Valuation

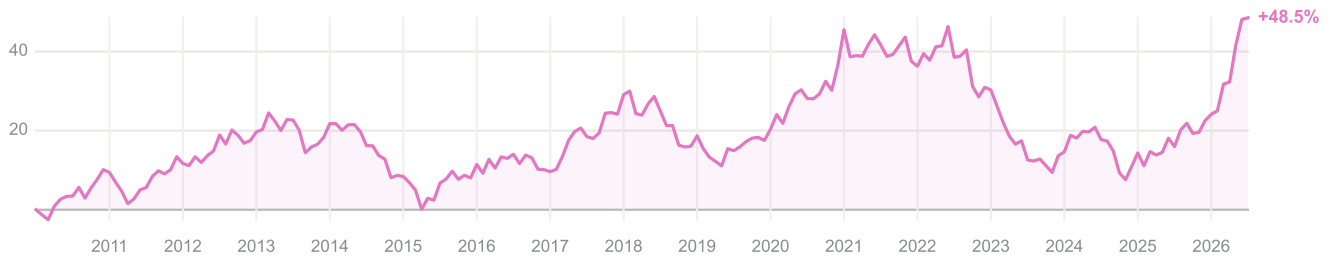
CAGR +2.4%



080 ● Operating Margin

Earnings Quality

CAGR +2.4%



081 ● 3Y Chg in Operating Cash Flow to Price

Historical Growth

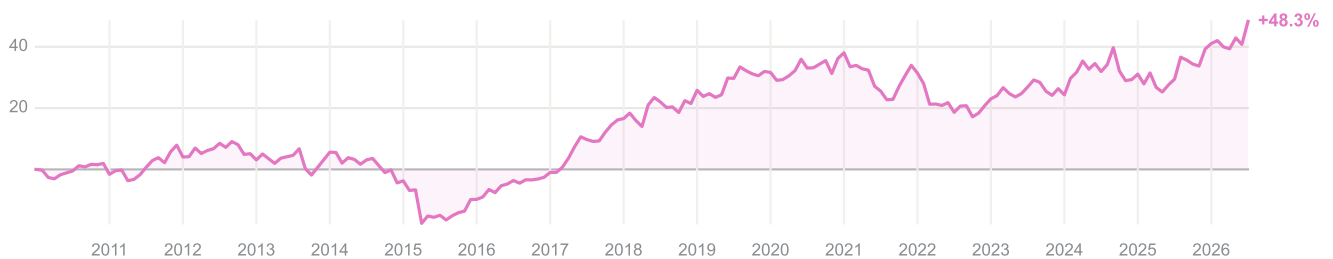
CAGR +2.4%



082 ● Retained Earnings to Total Assets

Earnings Quality

CAGR +2.4%



083 ● Inverse P/E Ratio Adj for Growth and Yield

Valuation

CAGR +2.4%

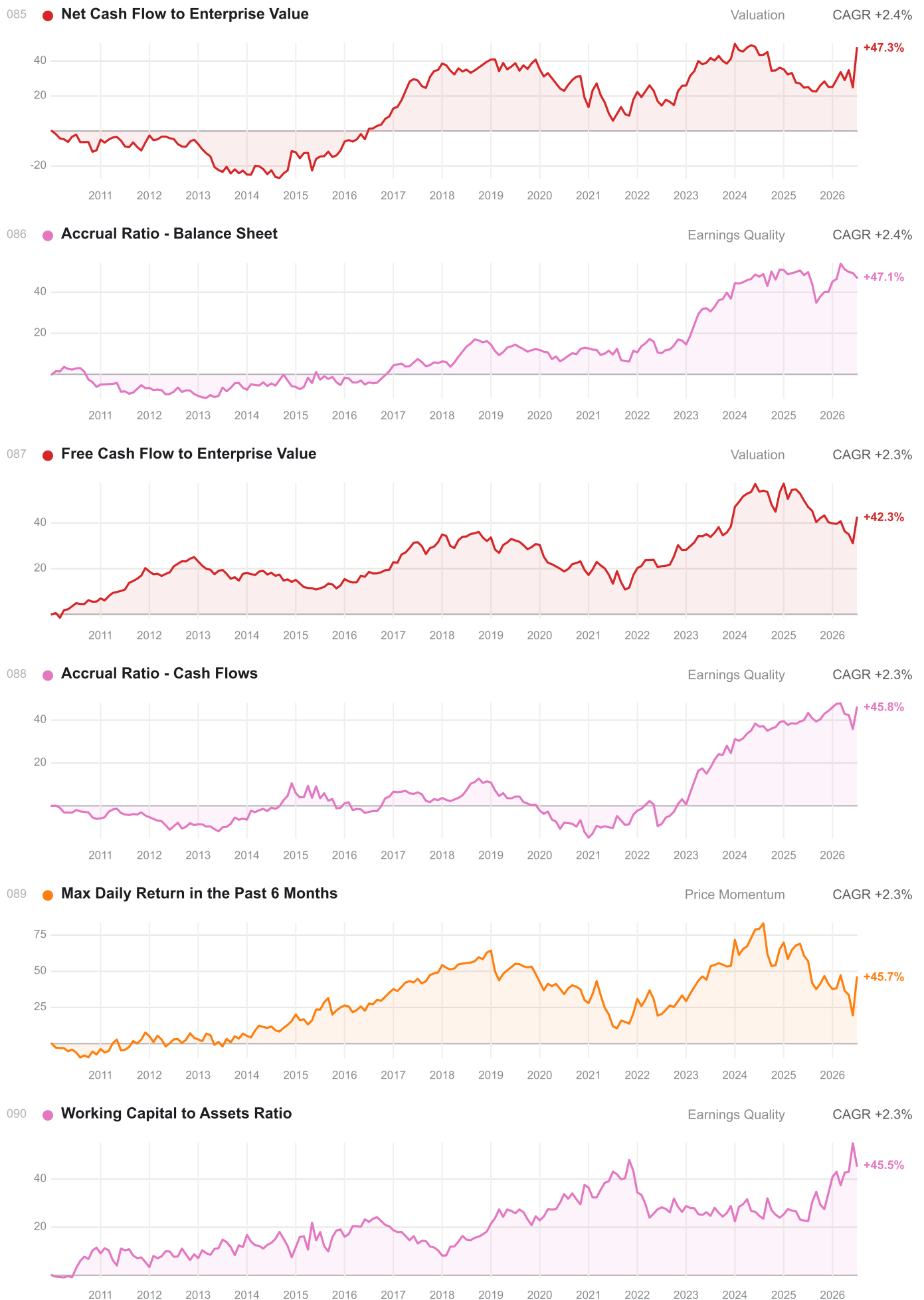


084 ● Cash to EV Ratio

Valuation

CAGR +2.4%





091 ● **Adjusted Forward Earnings to Price**

Valuation

CAGR +2.3%

092 ● **Log of Unadjusted Stock Price**

Price Momentum

CAGR +2.2%

093 ● **Book to Price**

Valuation

CAGR +2.2%

094 ● **Percent Above 260-Day Low**

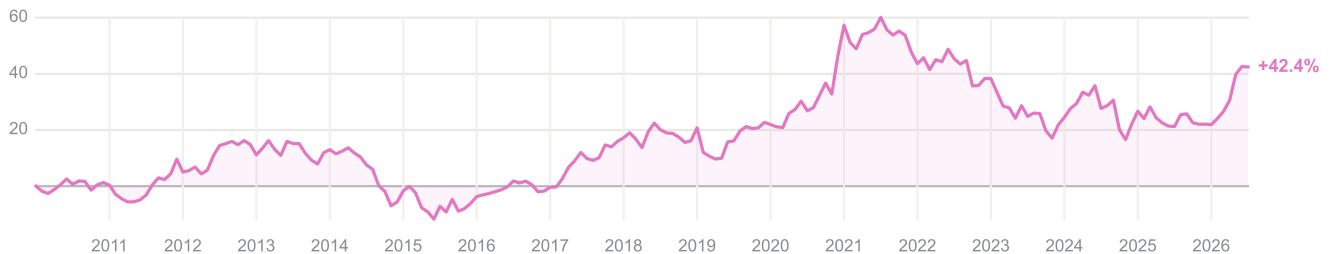
Price Momentum

CAGR +2.2%

095 ● **Pretax Income to Net Operating Assets**

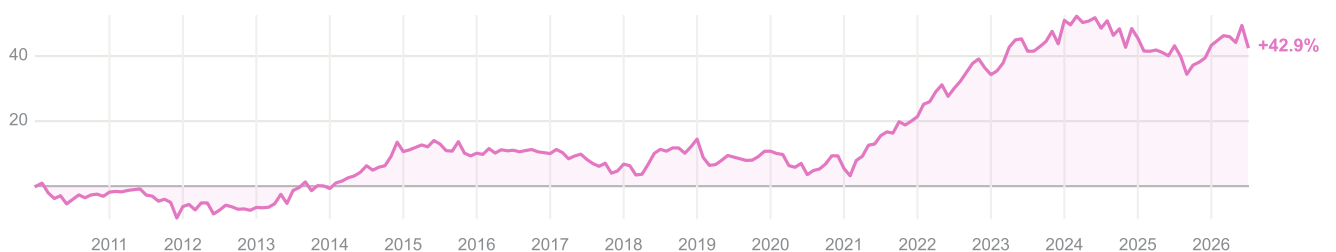
Earnings Quality

CAGR +2.2%

096 ● **Depreciation to Capex Ratio**

Earnings Quality

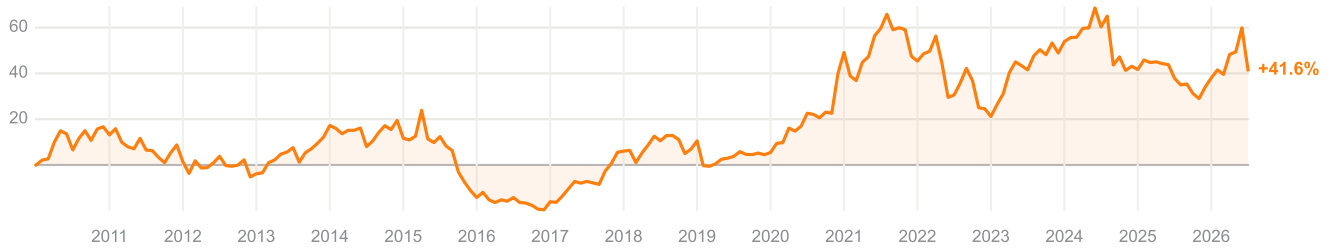
CAGR +2.1%



097 ● **15/36 Week Price Ratio**

Price Momentum

CAGR +2.1%

098 ● **Adjusted 6 Month Average Share Turn Over**

Price Momentum

CAGR +2.1%

099 ● **Solvency Ratio**

Valuation

CAGR +2.1%

100 ● **Cash to Total Assets**

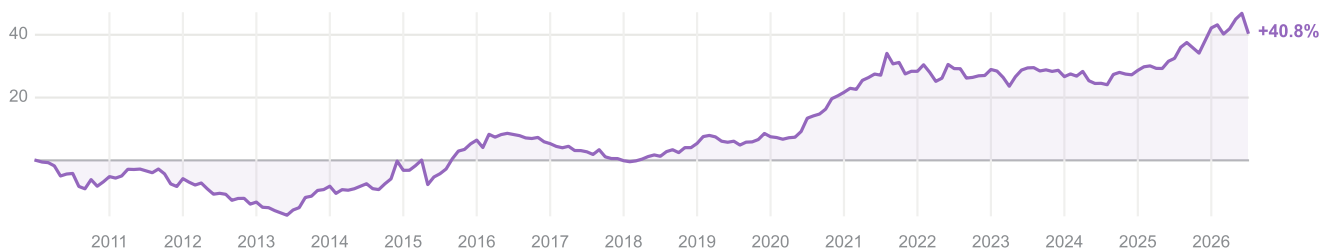
Valuation

CAGR +2.1%

101 ● **60M CAPM Beta**

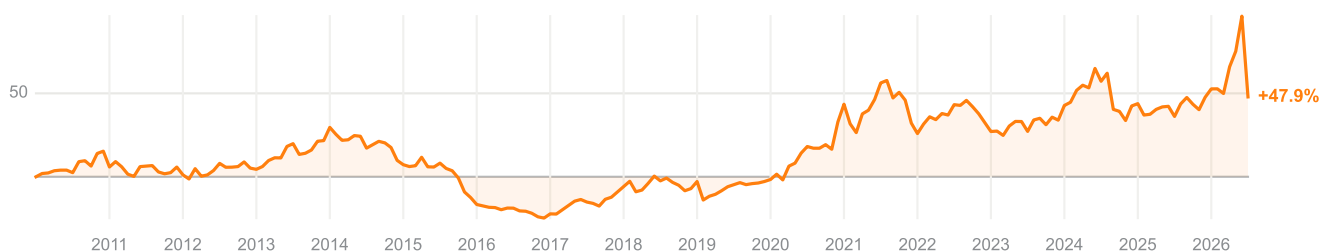
Volatility

CAGR +2.1%

102 ● **12M - 1M Price Momentum**

Price Momentum

CAGR +2.1%



103 ● Operating Earnings to Price Ratio

Valuation

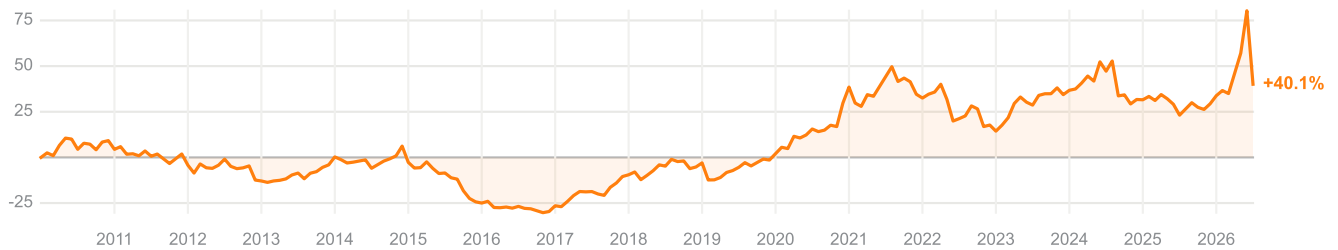
CAGR +2.1%



104 ● 50 Day to 200 Day Stock Price

Price Momentum

CAGR +2.1%



105 ● Interest Coverage Ratio

Capital Efficiency

CAGR +2.1%



106 ● Forward Free Cash Flow to Price Ratio

Valuation

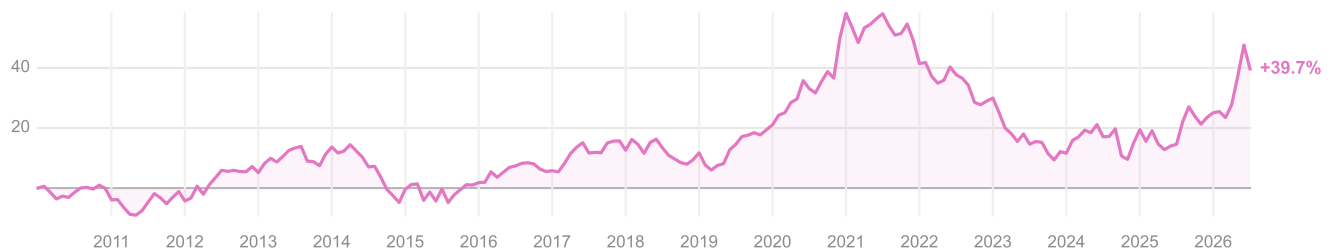
CAGR +2.0%



107 ● Cash Flow to Total Asset

Earnings Quality

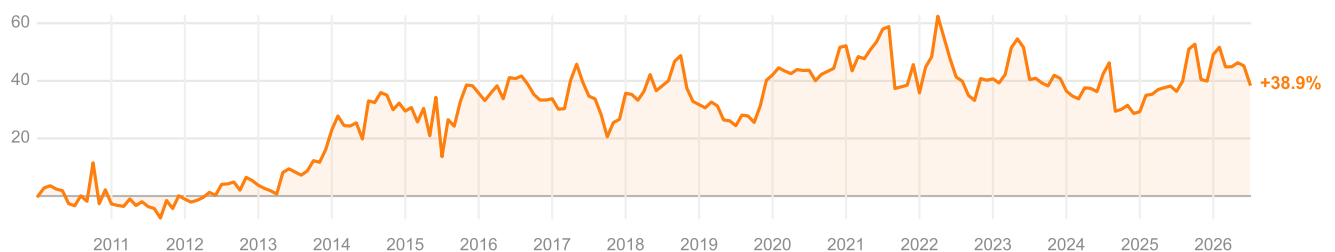
CAGR +2.0%

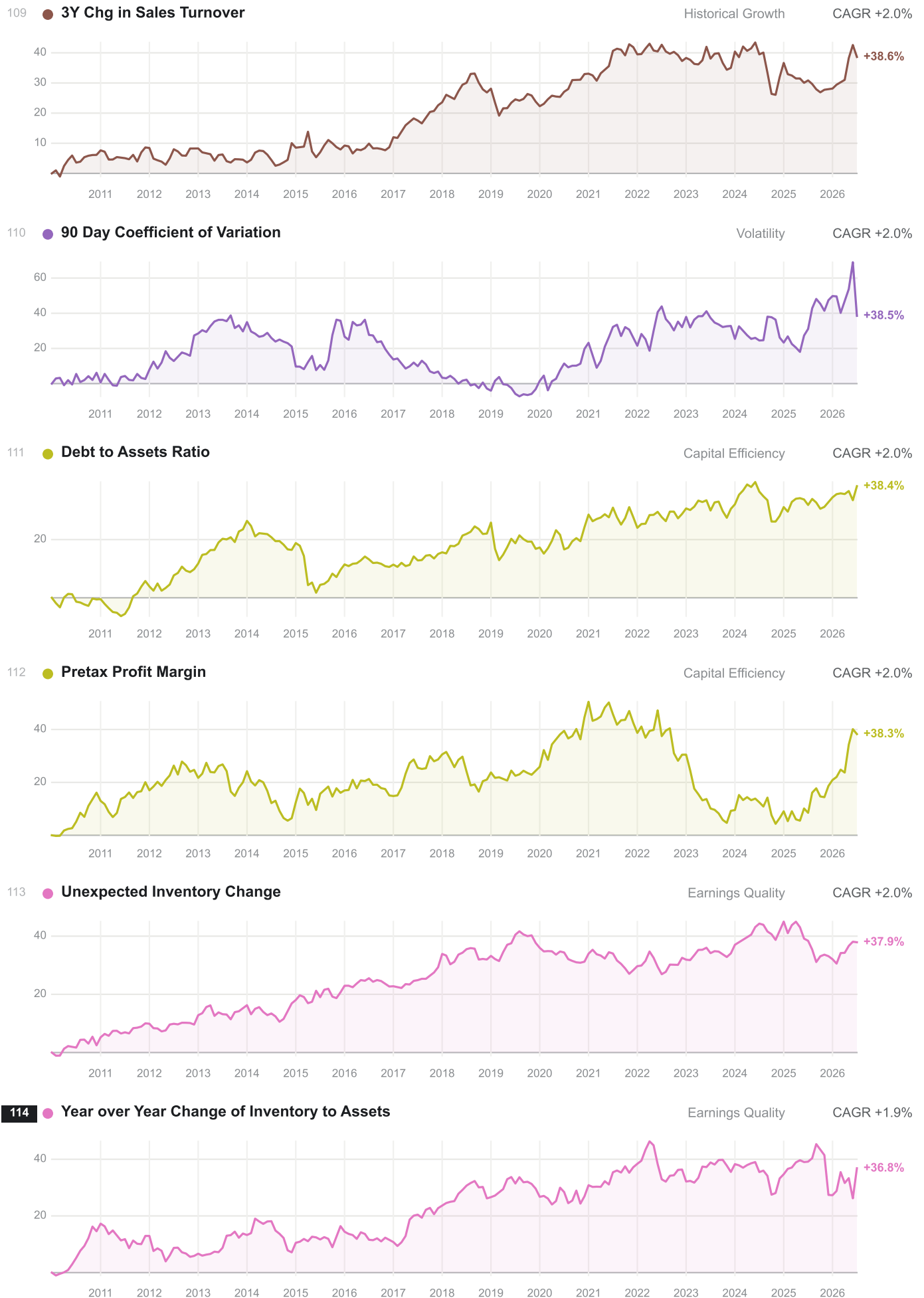


108 ● 50 Day Volume Signal

Price Momentum

CAGR +2.0%





115 ● **Institution Ownership Level**

Price Momentum

CAGR +1.9%

116 ● **Dividend Growth**

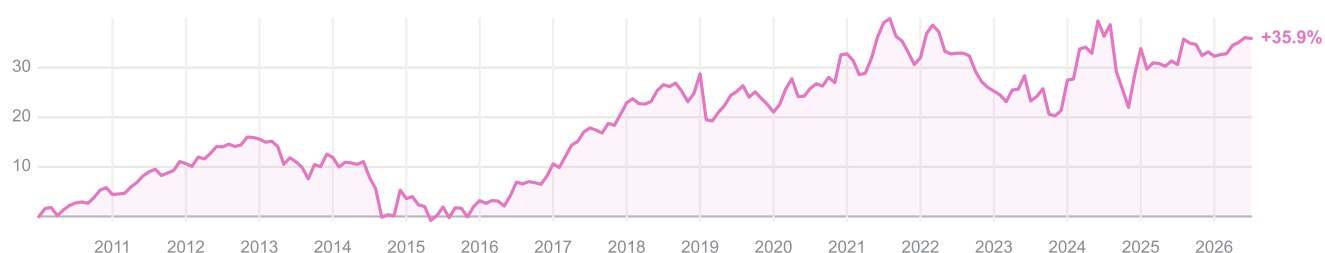
Historical Growth

CAGR +1.9%

117 ● **Year over Year Change of EPS to Sales**

Earnings Quality

CAGR +1.9%

118 ● **Change in TTM Depr. to Capex**

Earnings Quality

CAGR +1.9%

119 ● **3Y Chg in Asset Adjusted EPS**

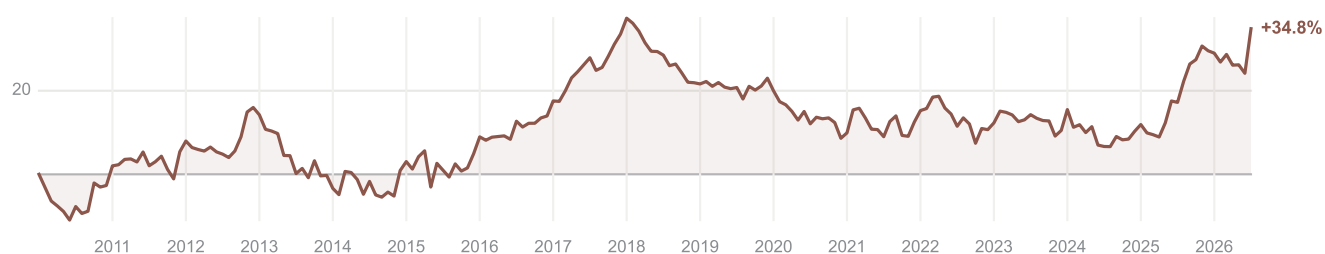
Historical Growth

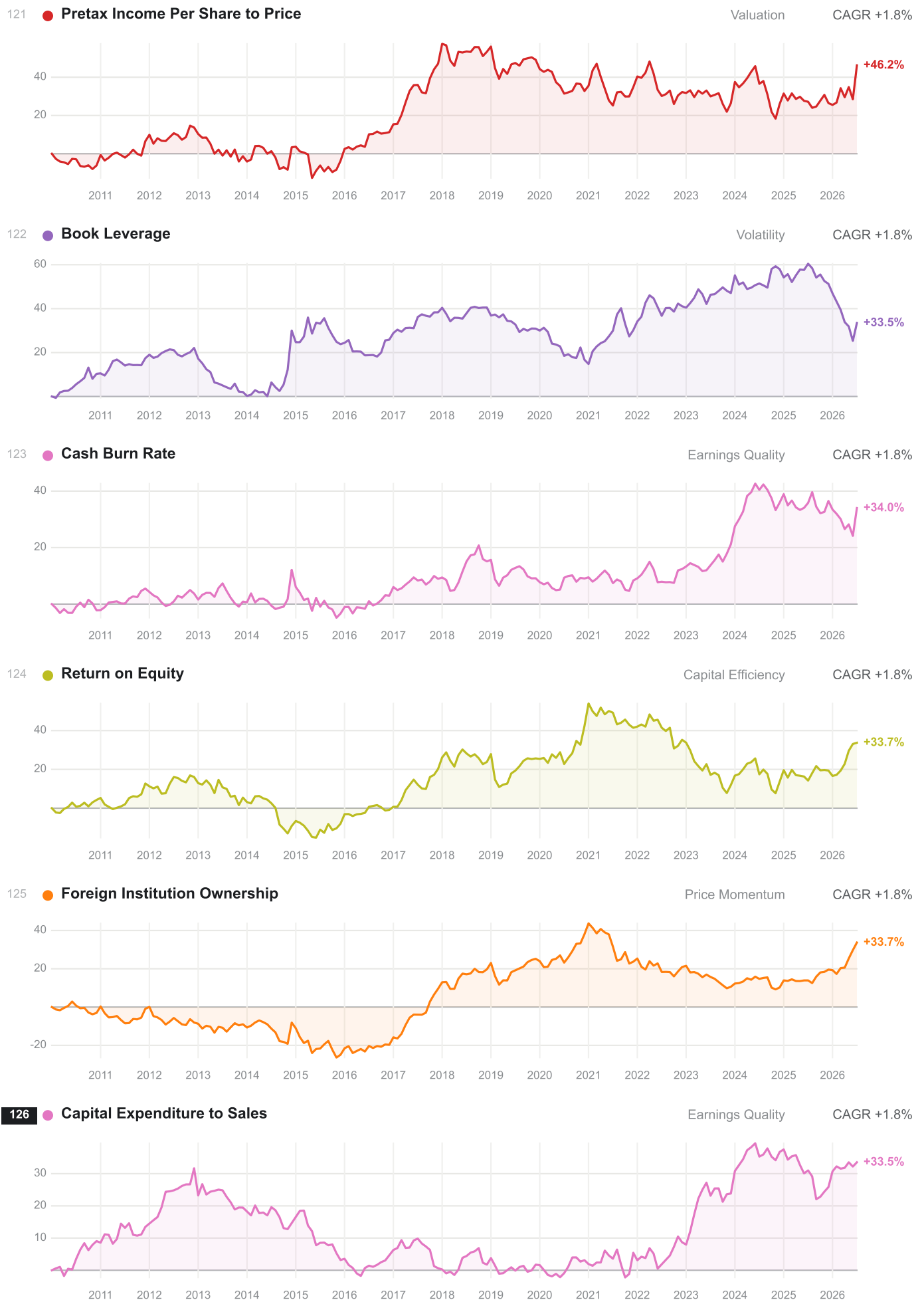
CAGR +1.8%

120 ● **3Y Chg in Cash Flow to Price**

Historical Growth

CAGR +1.8%





127 ● **Cash Flow to Invested Capital**

Earnings Quality

CAGR +1.8%

128 ● **EBITDA to Assets**

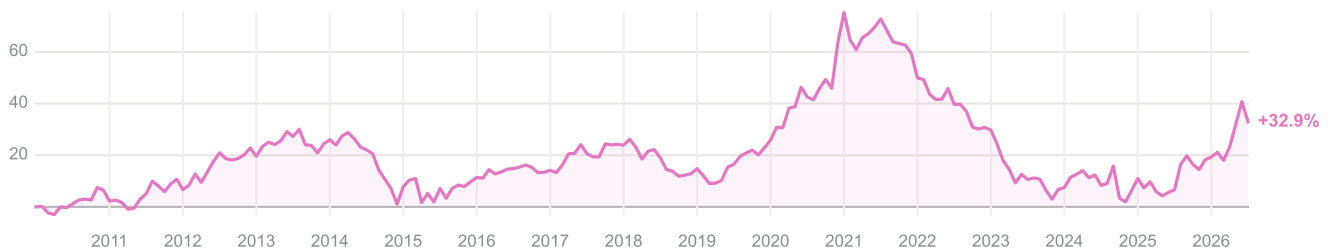
Capital Efficiency

CAGR +1.7%

129 ● **Operating Earnings to Assets Ratio**

Earnings Quality

CAGR +1.7%

130 ● **Return on Assets**

Capital Efficiency

CAGR +1.7%

131 ● **3Y Average Annual Sales Growth**

Historical Growth

CAGR +1.7%

132 ● **Net Current Assets to Price Ratio**

Valuation

CAGR +1.7%



133 ● **Institution Ownership Level - Active Manager**

Price Momentum

CAGR +1.7%

134 ● **Log of Market Capitalization Cubed**

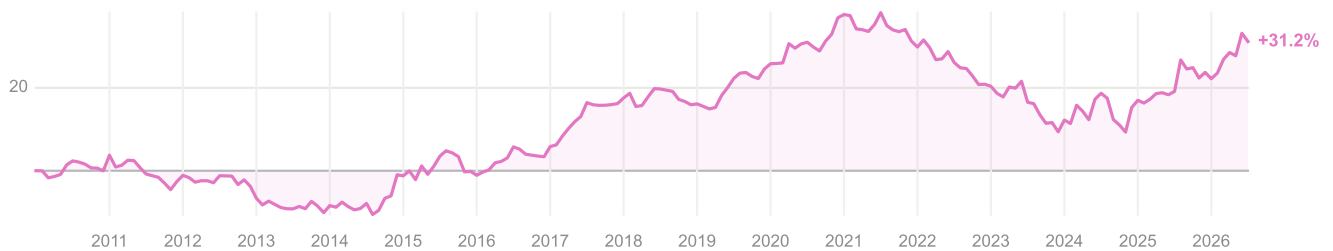
Size

CAGR +1.7%

135 ● **Year over Year Change in SGA to Sales**

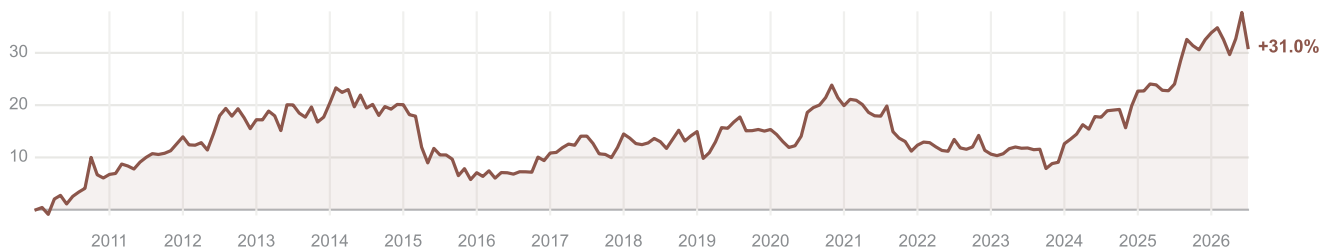
Earnings Quality

CAGR +1.7%

136 ● **3Y Chg in Free Cash Flow to Price**

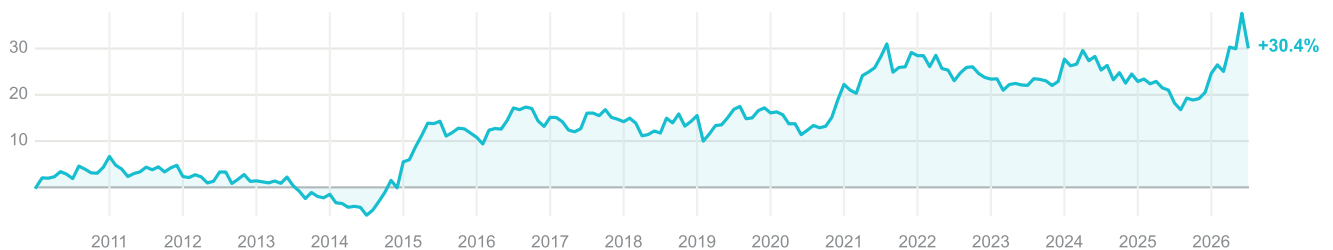
Historical Growth

CAGR +1.7%

137 ● **Buy-to-Sell Recommendation Ratio Less 3M SMA**

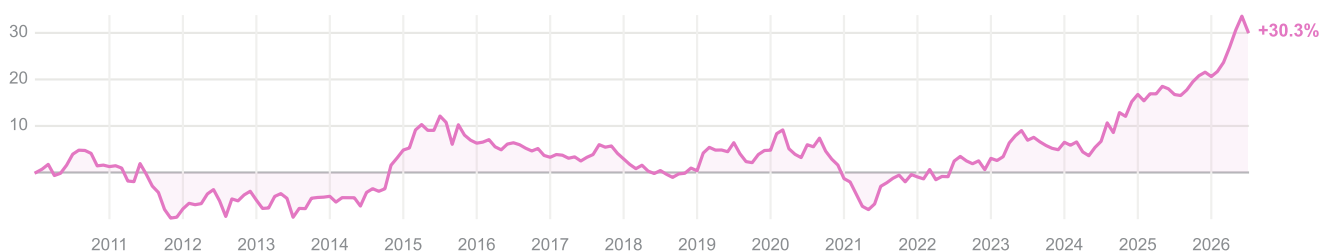
Analyst Expectations

CAGR +1.6%

138 ● **SG&A to Sales**

Earnings Quality

CAGR +1.6%



139 ● 3Y Chg in Asset Adjusted Free Cash Flow

Historical Growth

CAGR +1.6%



140 ● Working Capital to Total Assets

Earnings Quality

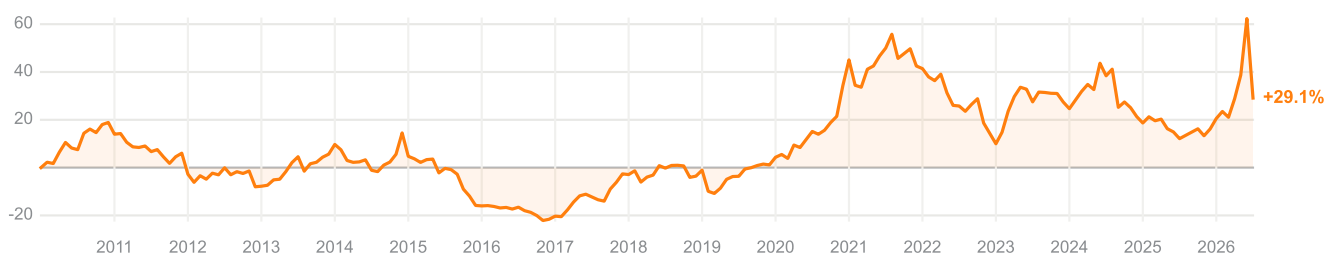
CAGR +1.6%



141 ● 26W Price Stat

Price Momentum

CAGR +1.6%



142 ● Dividends to Price Ratio

Valuation

CAGR +1.5%



143 ● Return on Net Tangible Assets

Valuation

CAGR +1.5%

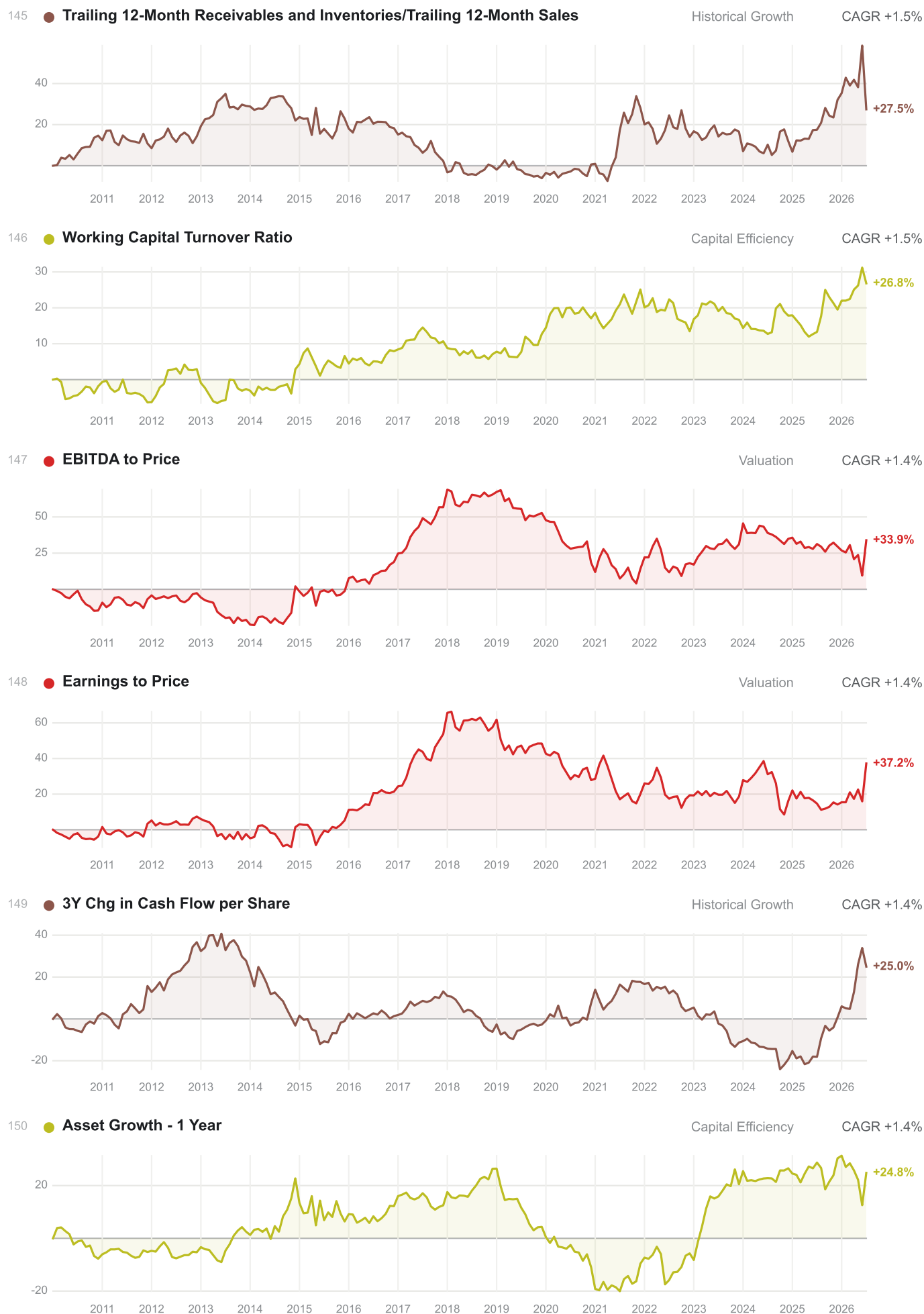


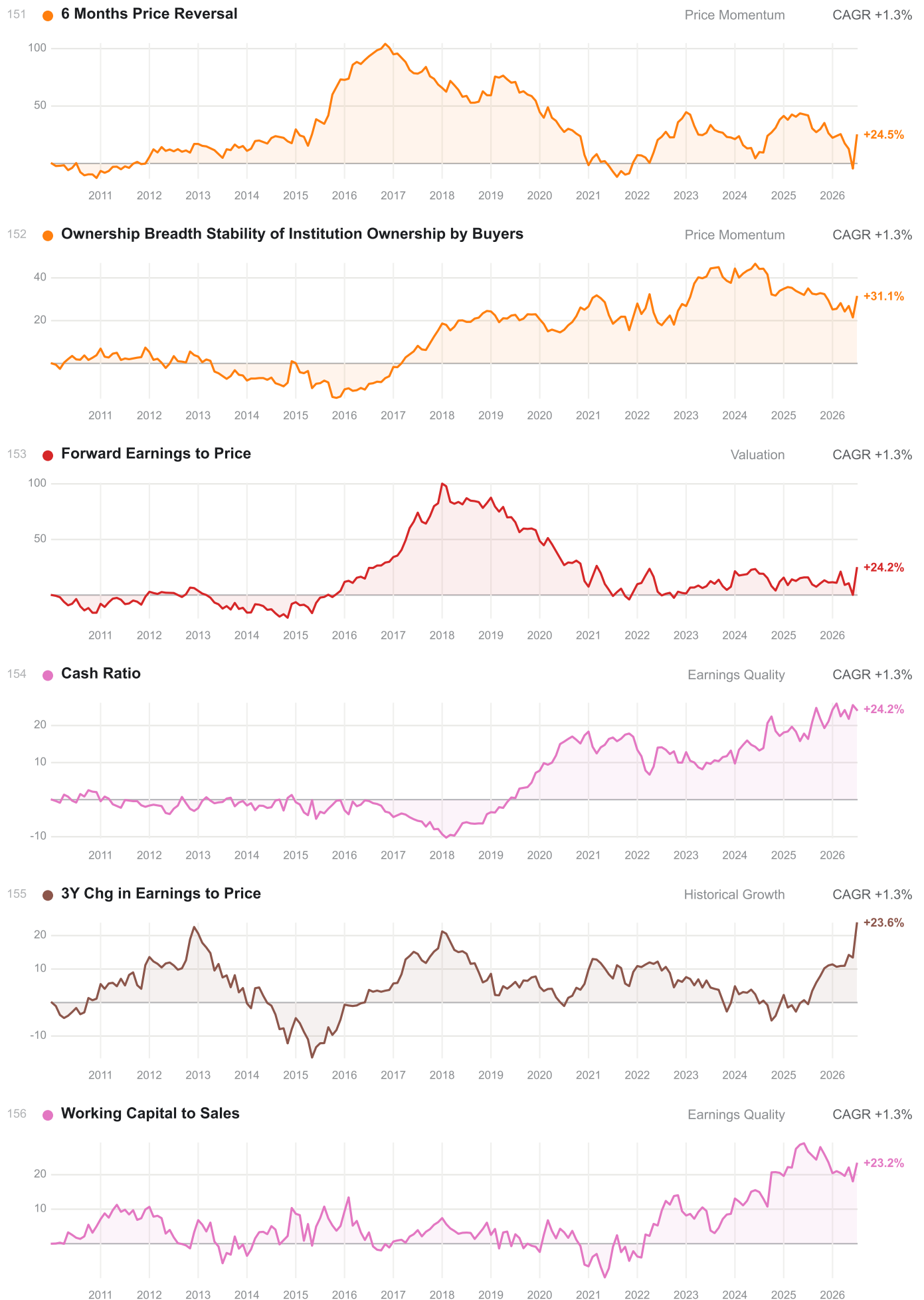
144 ● Log Market Cap

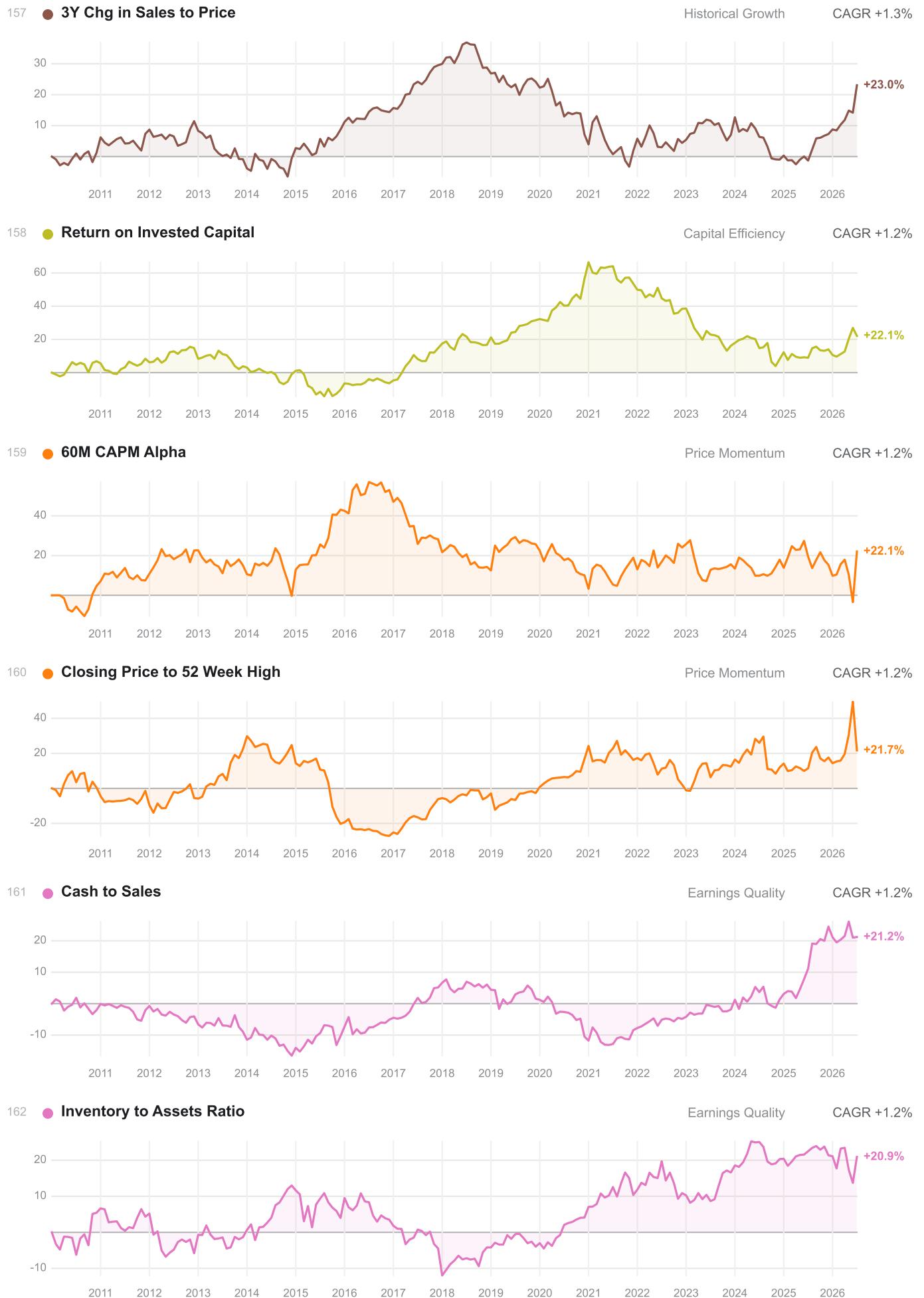
Size

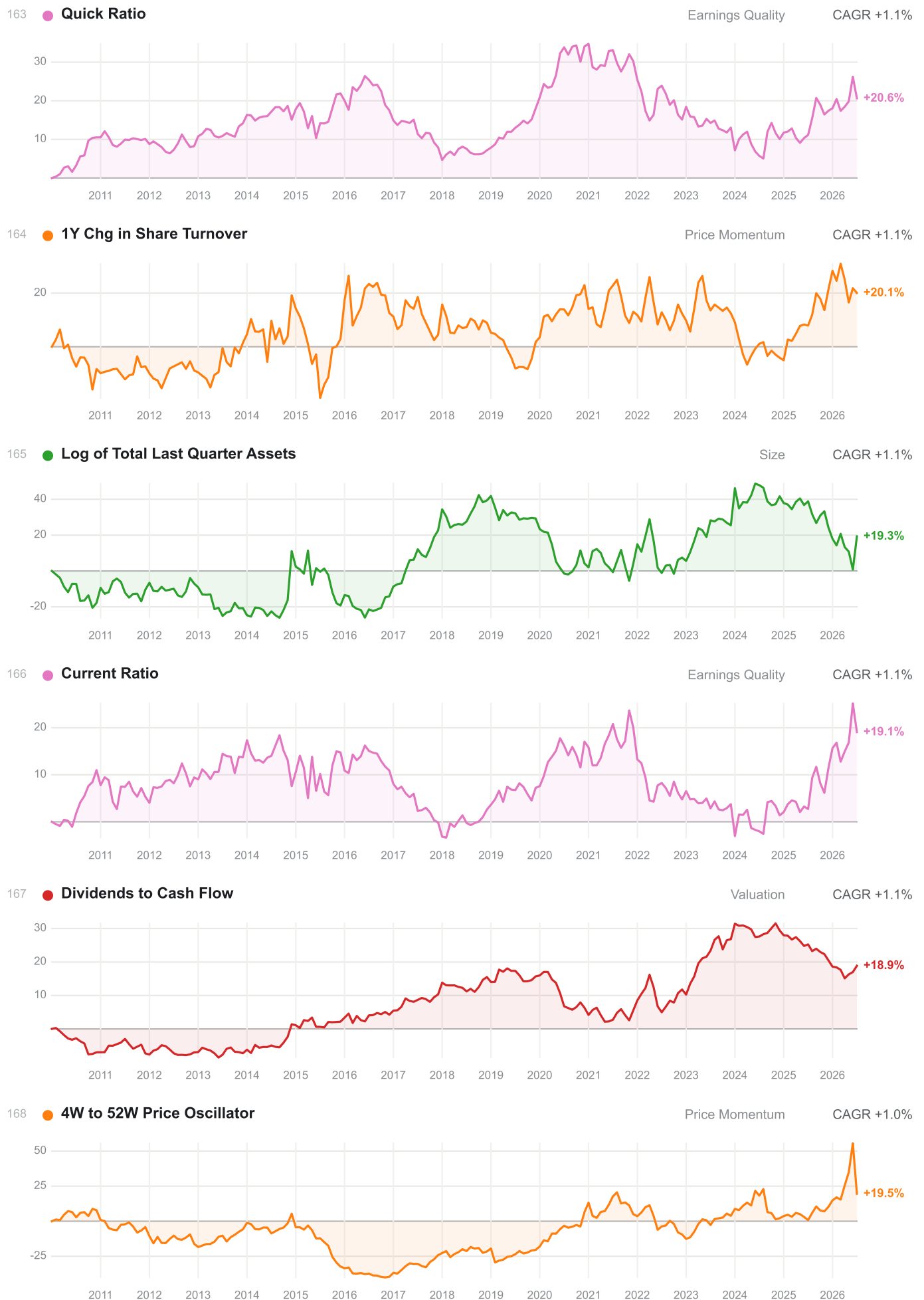
CAGR +1.5%











169 ● **Adj Number of EPS FY2 Revisions**

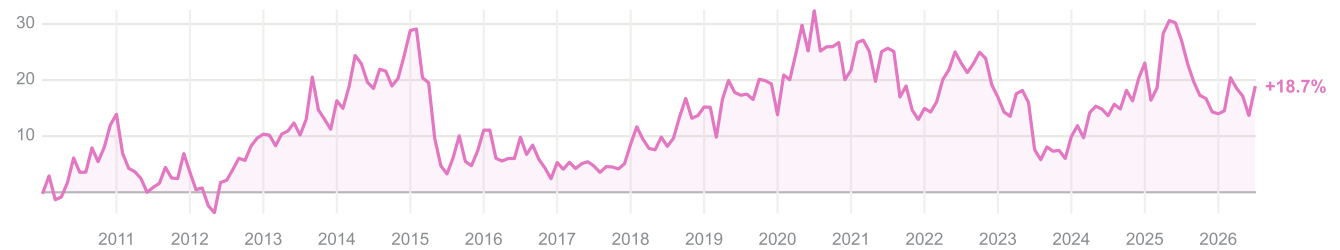
Analyst Expectations

CAGR +1.0%

170 ● **Inventory Turnover**

Earnings Quality

CAGR +1.0%

171 ● **6 Month Average Share Turn Over**

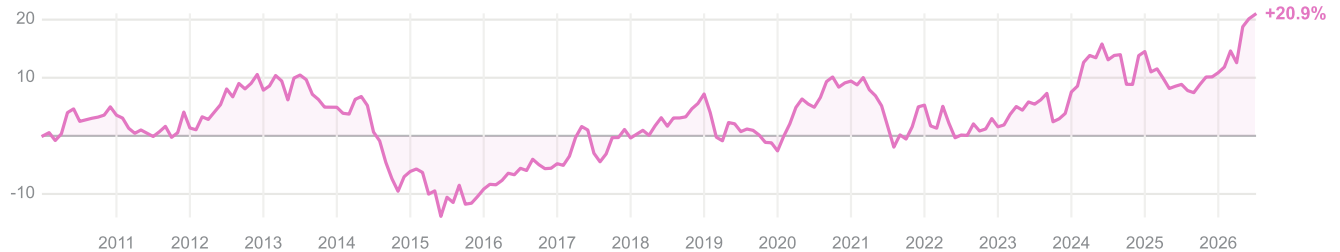
Price Momentum

CAGR +1.0%

172 ● **Change of Sales to Change of EPS**

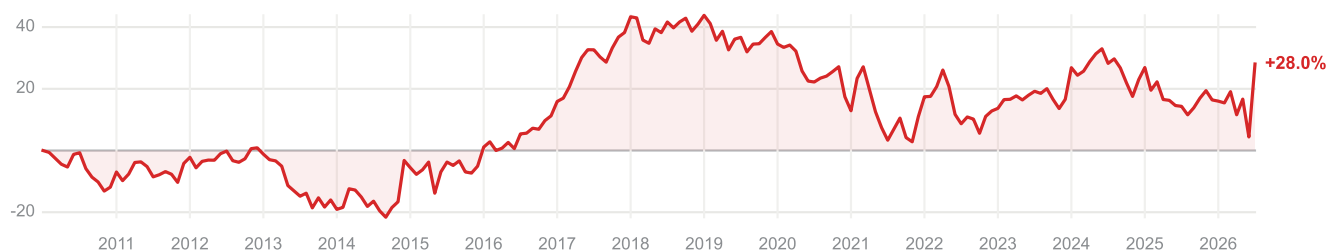
Earnings Quality

CAGR +1.0%

173 ● **EBITDA to Enterprise Value**

Valuation

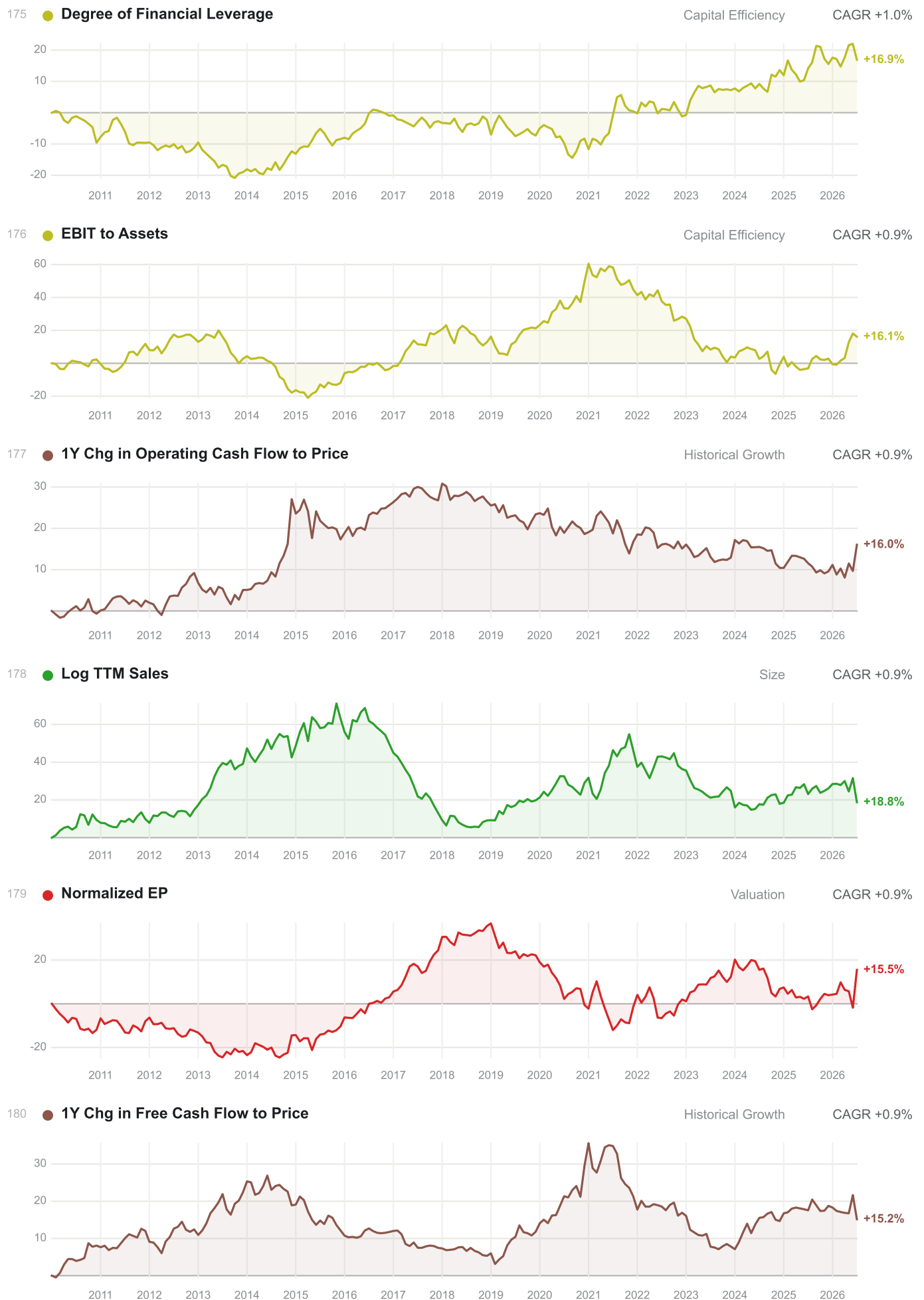
CAGR +1.0%

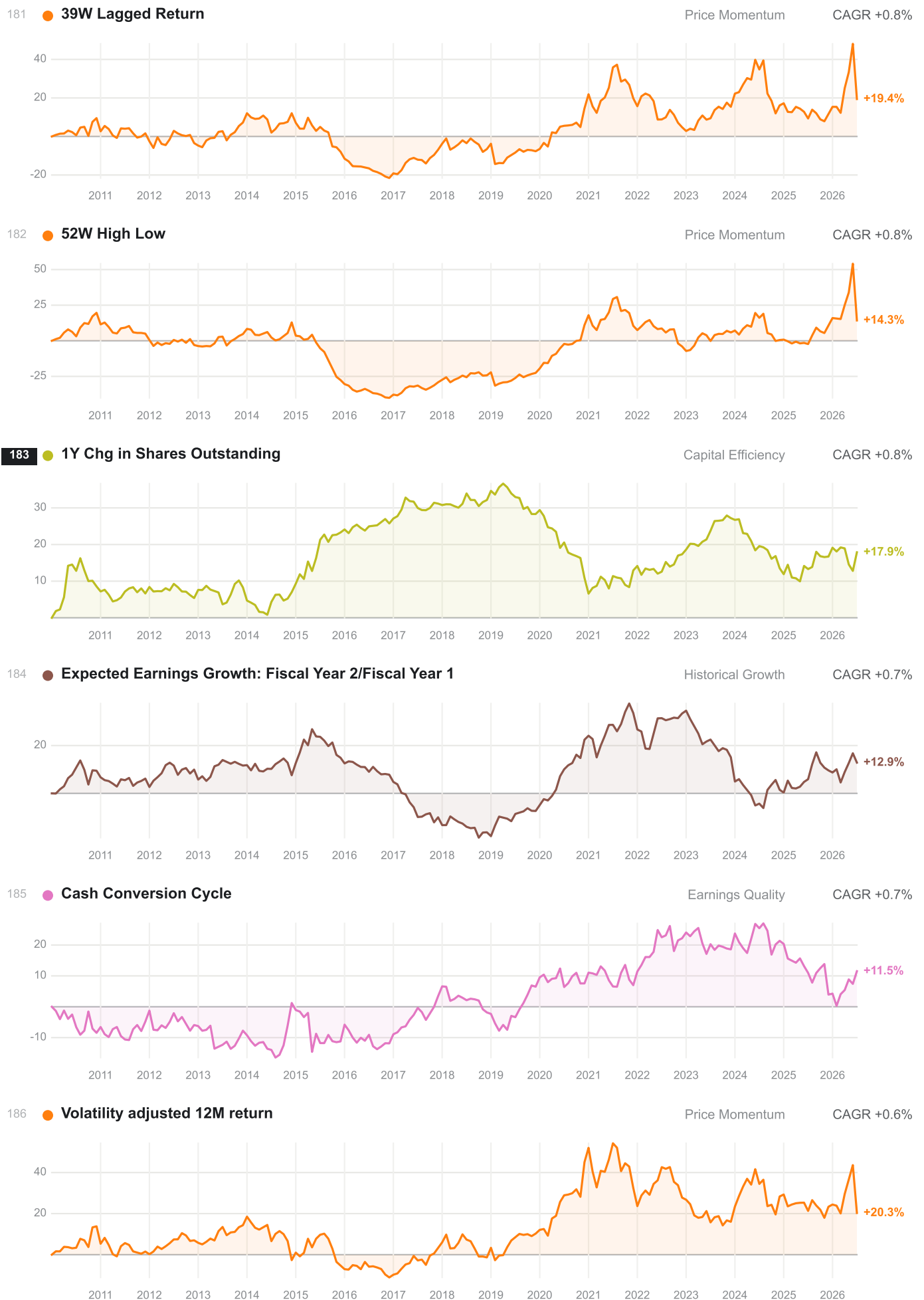
174 ● **8-Week Change in 12-Month Forward Earnings Consensus Estimate/Price**

Valuation

CAGR +1.0%



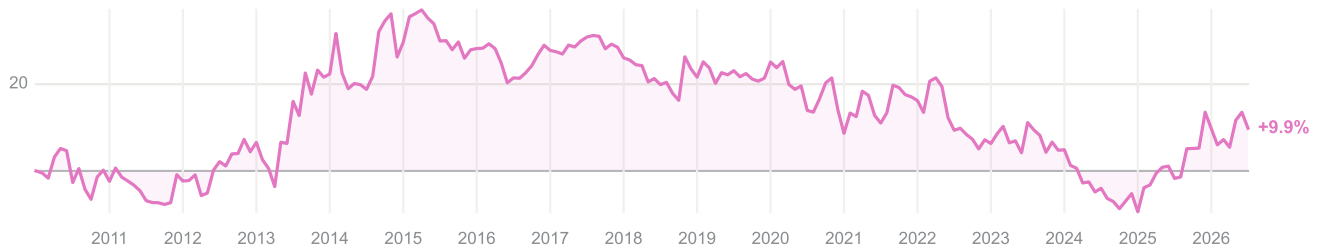




187 1Y Chg in Sales to Earnings

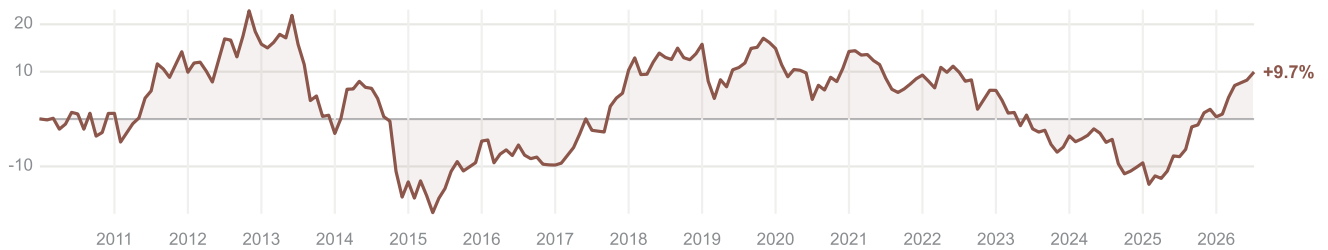
Earnings Quality

CAGR +0.6%

**188 Sustainable Growth Rate**

Historical Growth

CAGR +0.6%

**189 Year over Year Change of Total Debt**

Capital Efficiency

CAGR +0.5%

**190 Normalized ROE**

Capital Efficiency

CAGR +0.5%

**191 Capital Expenditure to Total Assets**

Capital Efficiency

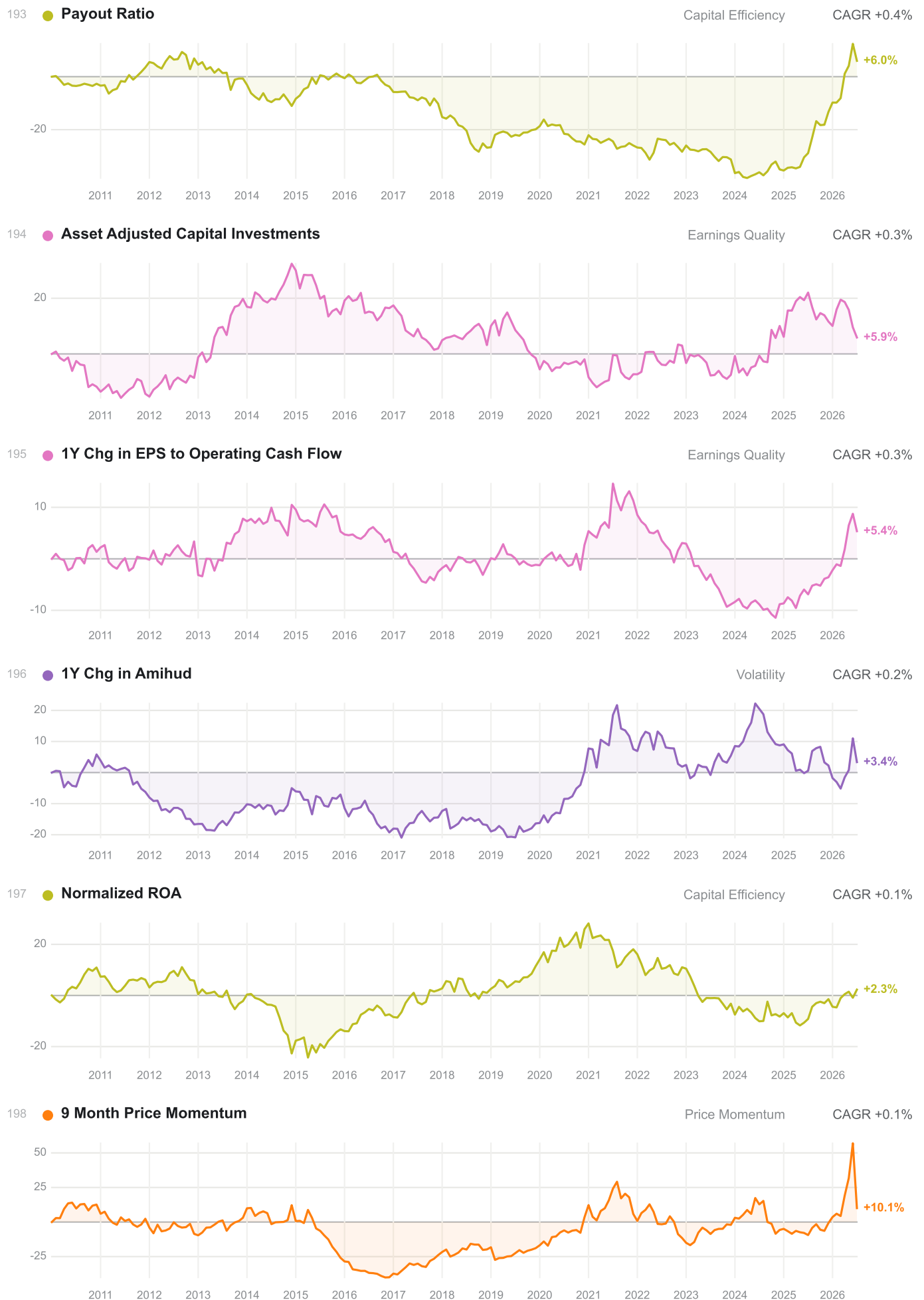
CAGR +0.5%

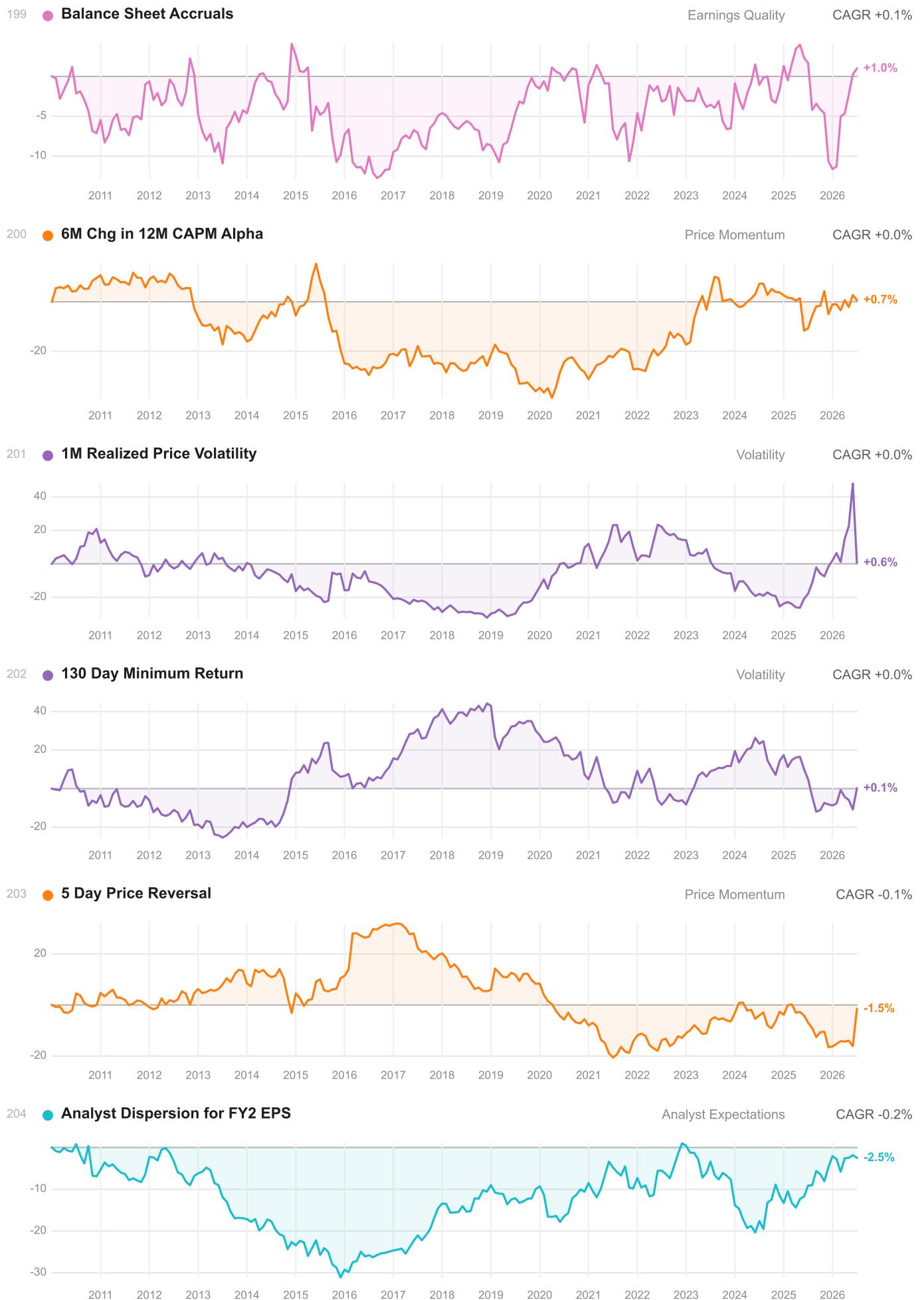
**192 Trailing 12-Month Overhead/Trailing 12-Month Sales**

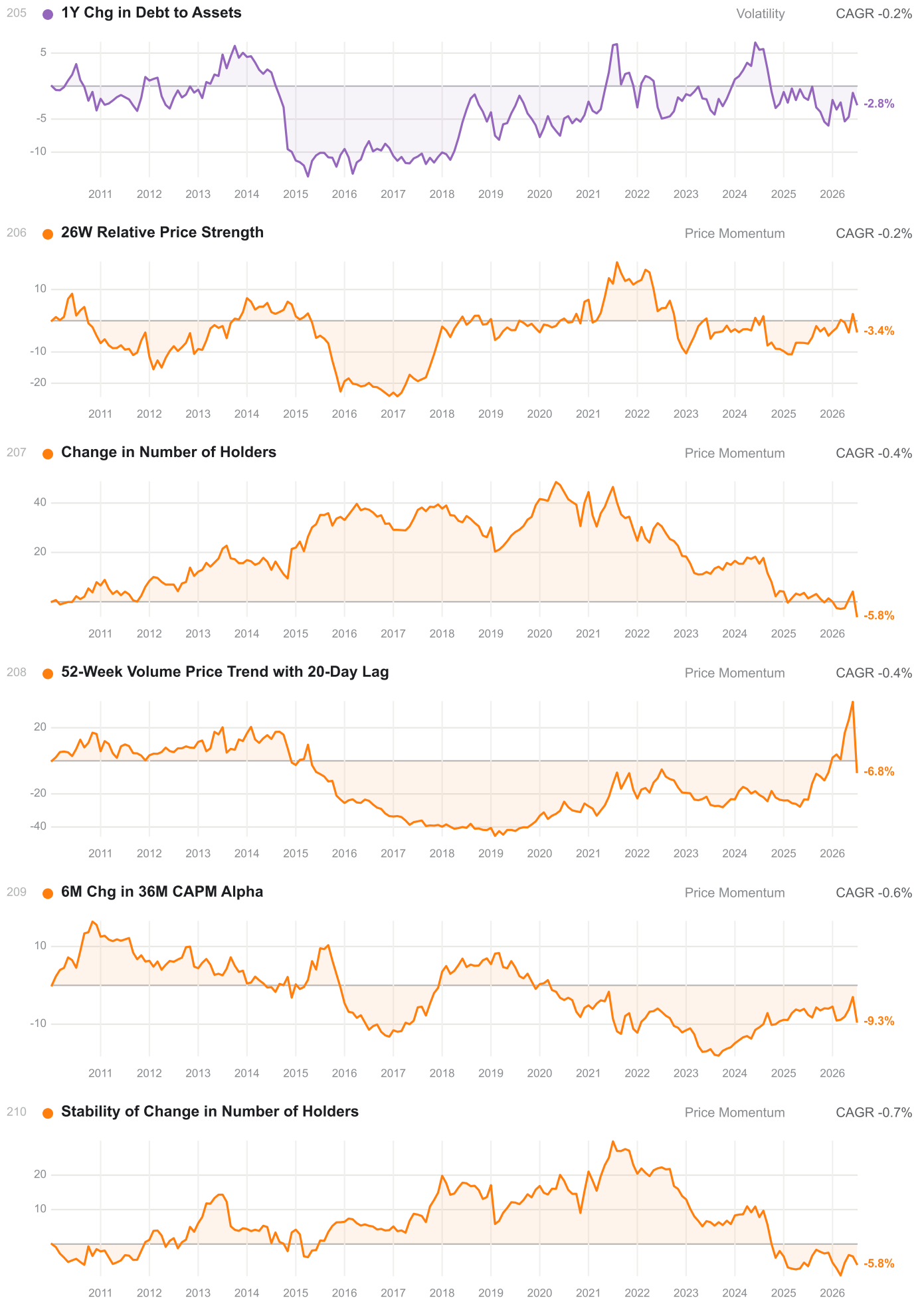
Historical Growth

CAGR +0.4%





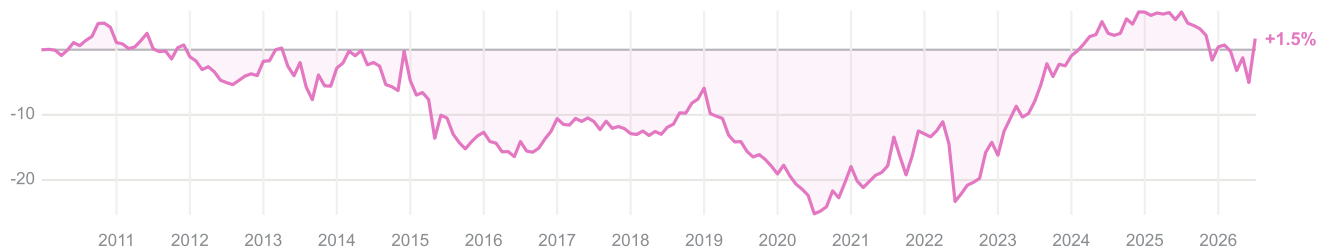




211 ● Sales to Gross Profit Margin Growth

Earnings Quality

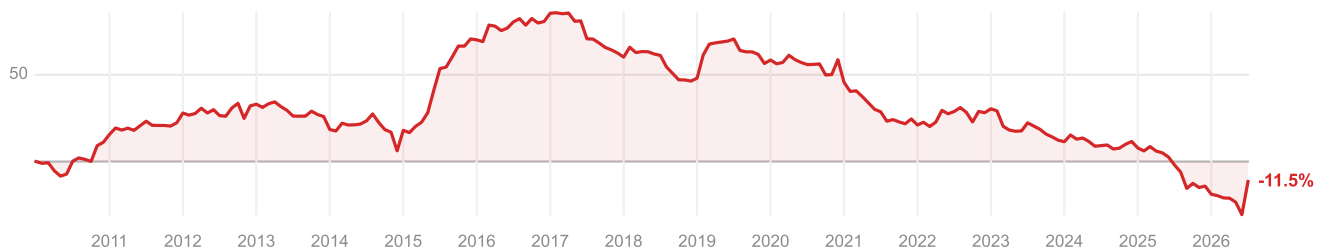
CAGR -0.7%



212 ● 4-Week Change in 12-Month Forward Earnings Consensus Estimate/Price

Valuation

CAGR -0.7%



213 ● Institution Ownership Turnover

Price Momentum

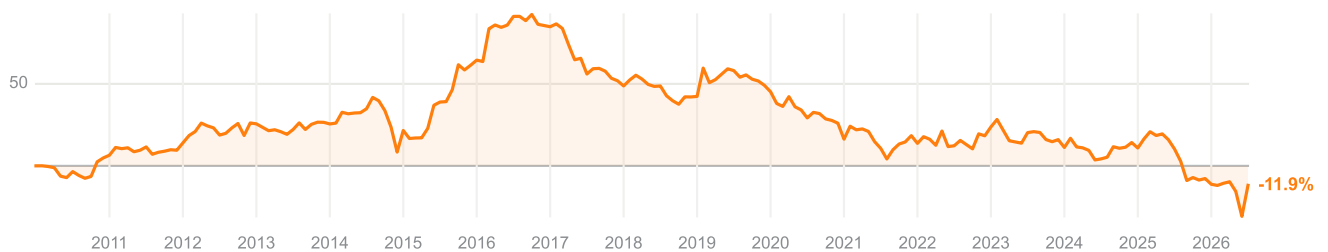
CAGR -0.8%



214 ● 1M Price Reversal

Price Momentum

CAGR -0.8%



215 ● Revenue Estimate Dispersion

Analyst Expectations

CAGR -1.0%



216 ● 14-Day Relative Strength Index

Price Momentum

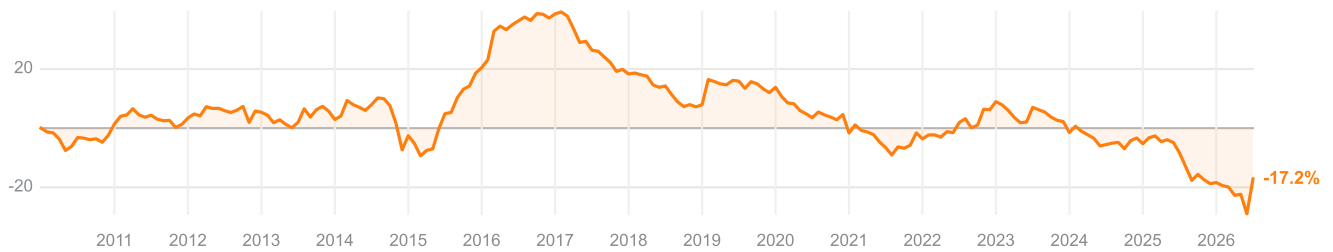
CAGR -1.1%



217 ● 1M Price High - 1M Price Low

Price Momentum

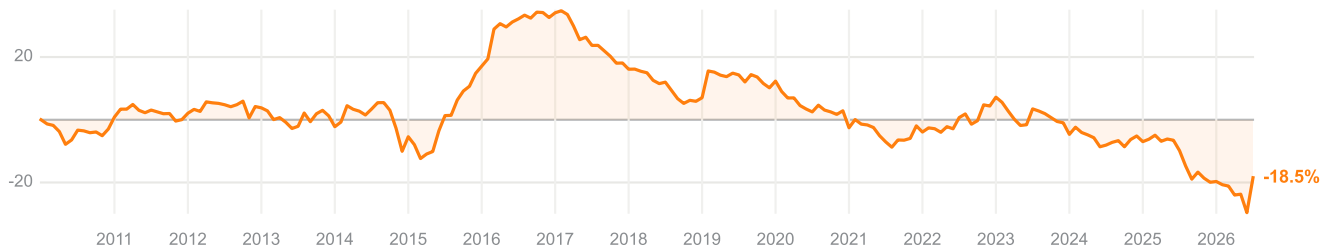
CAGR -1.1%



218 ● 20-Day Lane's Stochastic Indicator

Price Momentum

CAGR -1.2%



219 ● EBIT Estimate Dispersion

Analyst Expectations

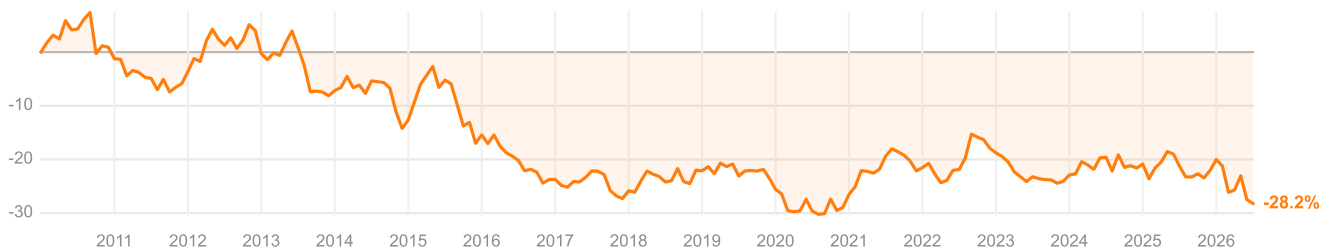
CAGR -1.7%



220 ● 6M Chg in 18M CAPM Alpha

Price Momentum

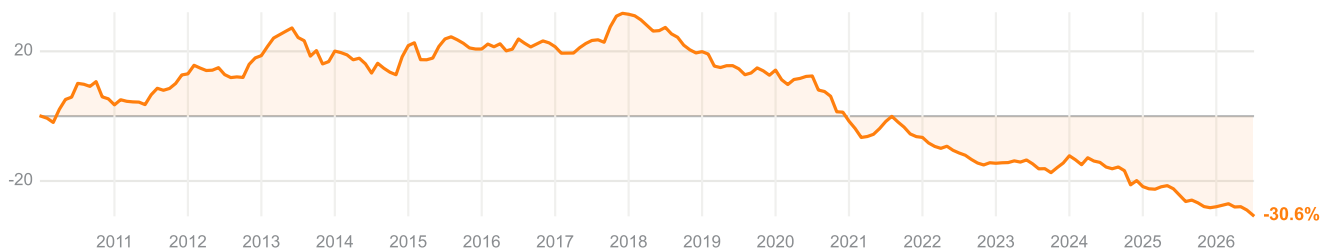
CAGR -2.0%



221 ● Change of Ownership Level - All Managers

Price Momentum

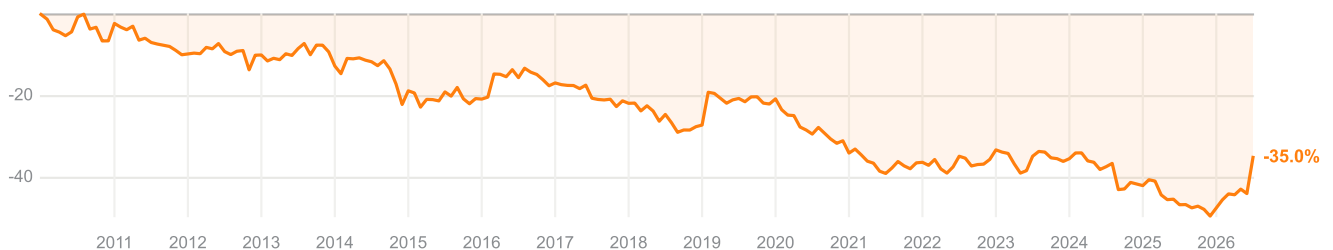
CAGR -2.2%



222 ● 5-Day Money Flow/Volume

Price Momentum

CAGR -2.6%



223 ● 5 Day Volume Signal

Price Momentum

CAGR -4.5%



224 ● 20 Day Coefficient of Variation of Volume to Price

Price Momentum

CAGR -5.6%



225 ● 30 Day Coefficient of Variation of Volume to Price

Price Momentum

CAGR -6.0%

